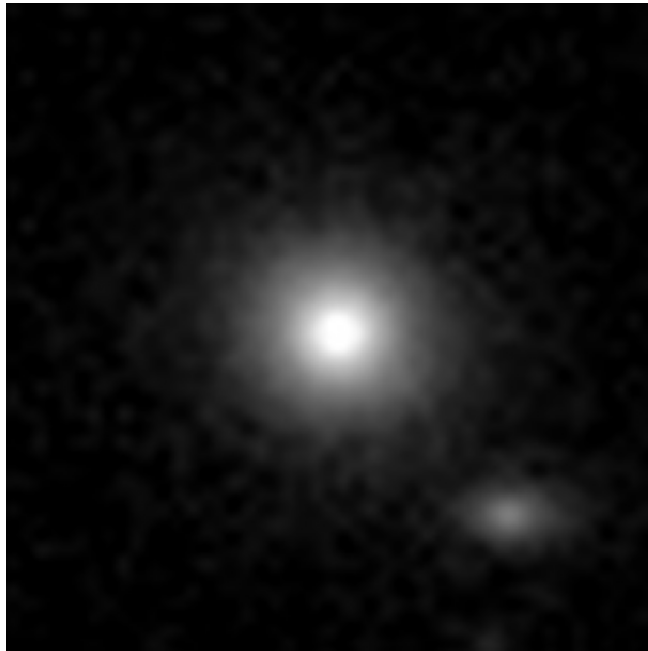
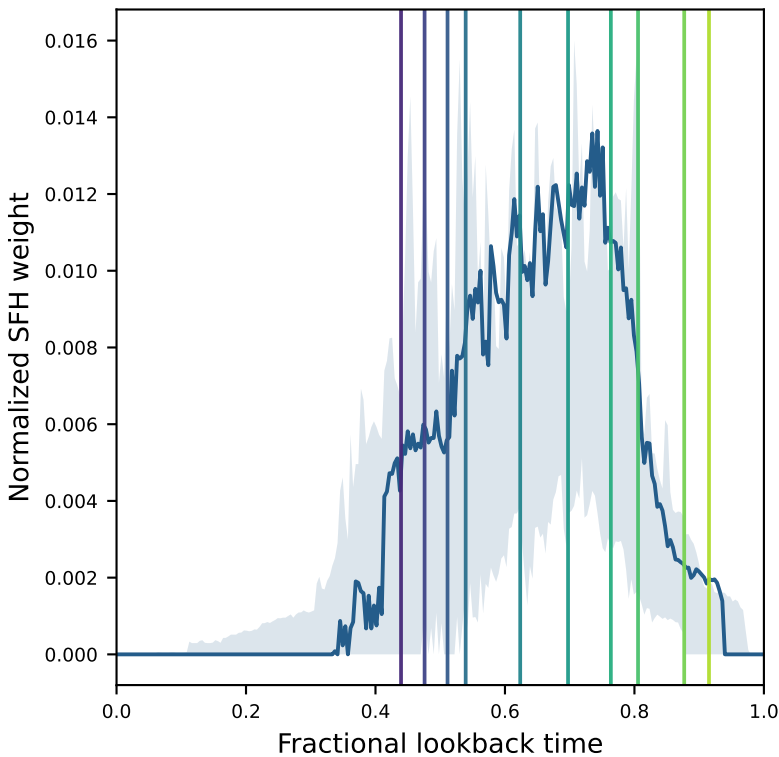


Pilot Quenched-descendant progenitor-analogue sequence

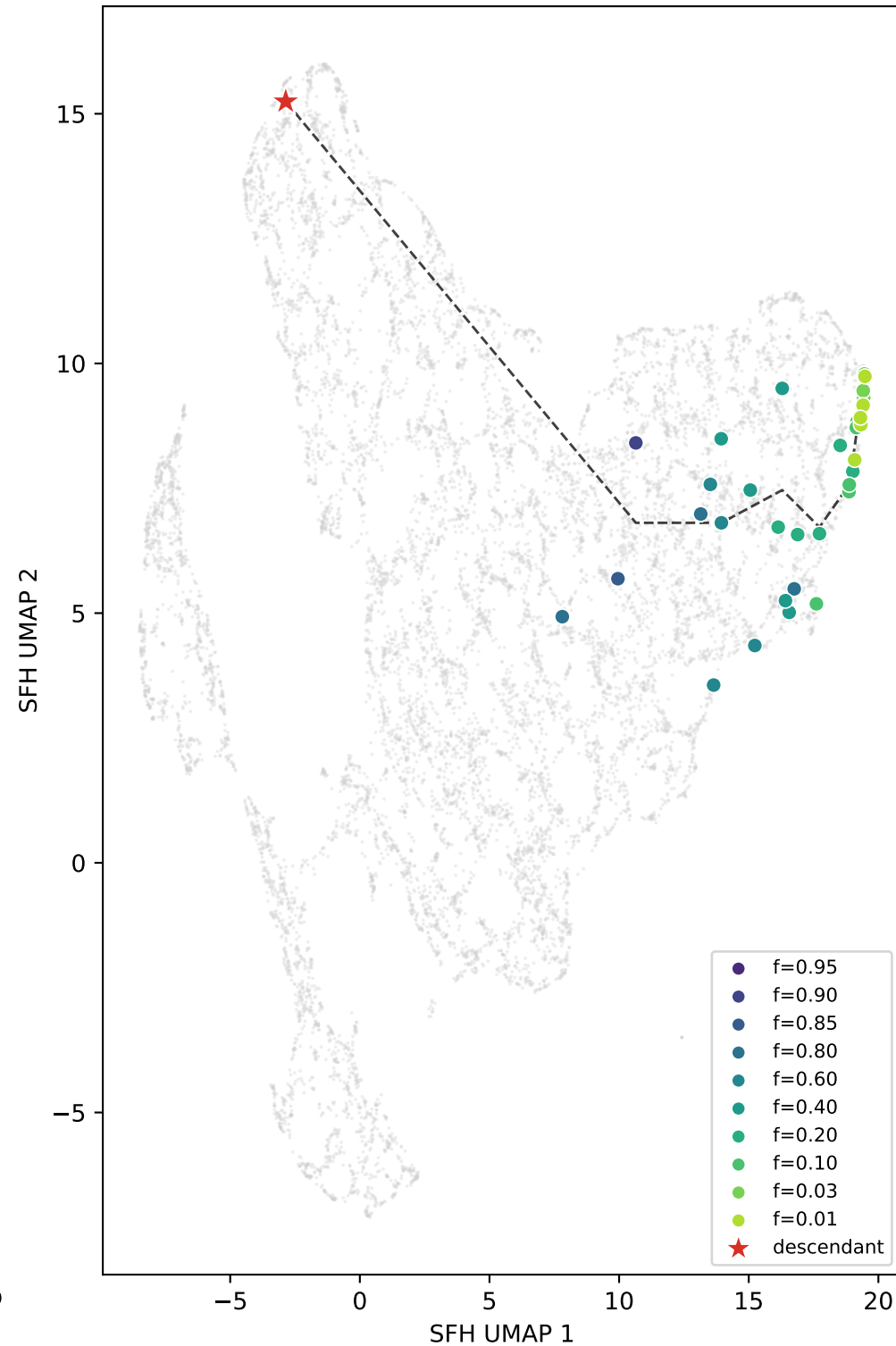
Descendant 2690824867644639038
log M*=11.00, z=0.794, log sSFR=-12.56



Descendant SFH and earlier-state checkpoints

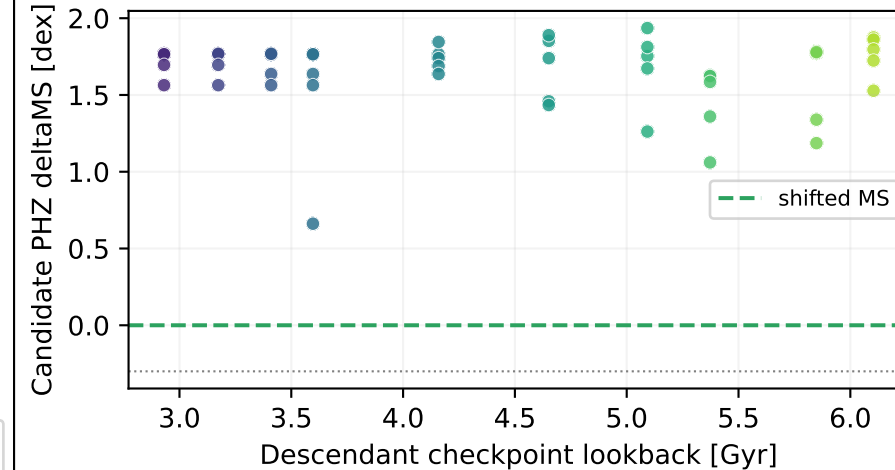


Analogue positions after SFH-only selection



f	L	logM	T	N	zmed
0.95	2.93	10.98	2.00	1,141	0.70
0.90	3.17	10.95	2.00	1,226	0.69
0.85	3.41	10.93	2.00	1,311	0.69
0.80	3.60	10.90	2.00	1,392	0.67
0.60	4.16	10.78	2.00	1,663	0.63
0.40	4.65	10.60	2.00	1,863	0.60
0.20	5.09	10.30	1.58	1,998	0.54
0.10	5.37	10.00	1.30	2,042	0.47
0.03	5.85	9.48	0.82	1,655	0.34
0.01	6.10	9.00	0.57	878	0.23

Each candidate evaluated at its own z and mass



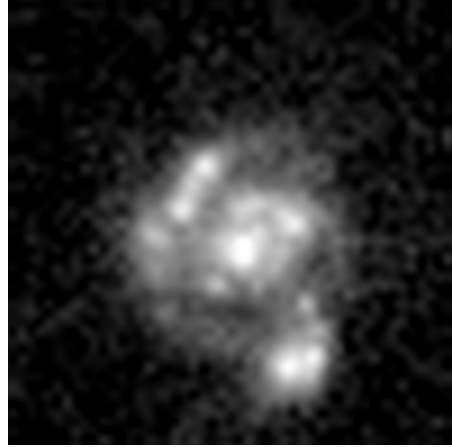
Selection variables

- predicted stellar mass, within +/-0.15 dex
- cumulative SFH before each descendant state
- 2 Gyr physical-time comparison when available

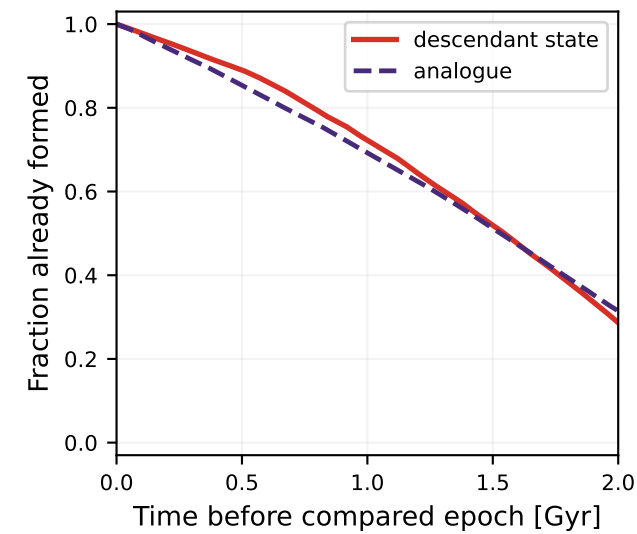
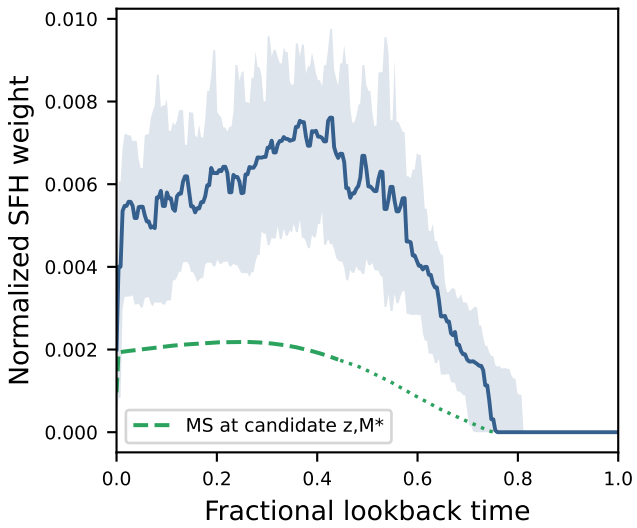
Candidate redshift, morphology, deltaMS, image embedding, and UMAP position are outcomes, not selection variables. The dashed path joins ensemble medians, not one galaxy's orbit.

Earlier state $f=0.95$: lookback=2.93 Gyr, predicted $\log M^*=10.98$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

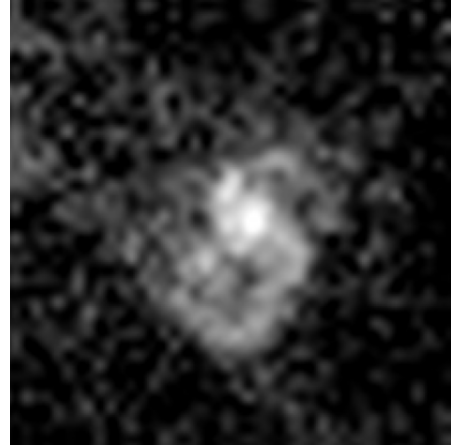
rank 1: 2659854720684798410
 $\log M^*=10.93$ (-0.05) $z=1.378$
 $\Delta MS=+1.77$ $D=0.020$



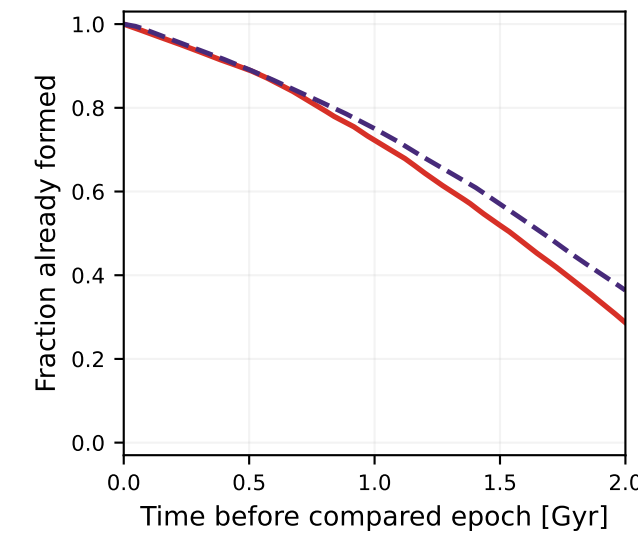
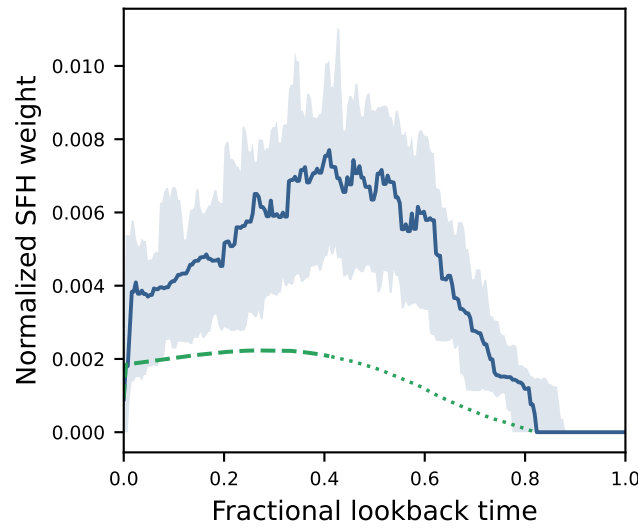
Complete candidate SFH



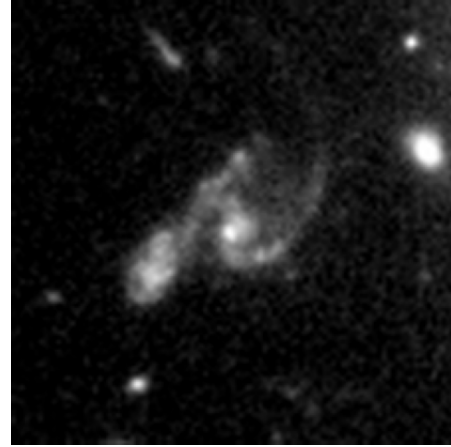
rank 2: 2753316360680891979
 $\log M^*=11.08$ (+0.10) $z=1.472$
 $\Delta MS=+1.77$ $D=0.029$



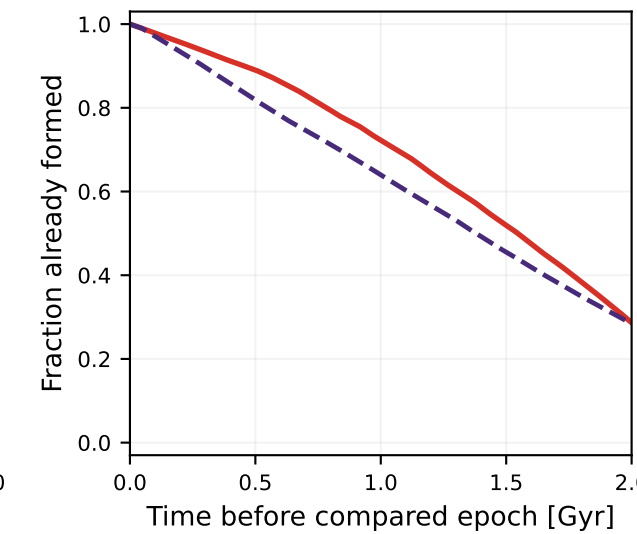
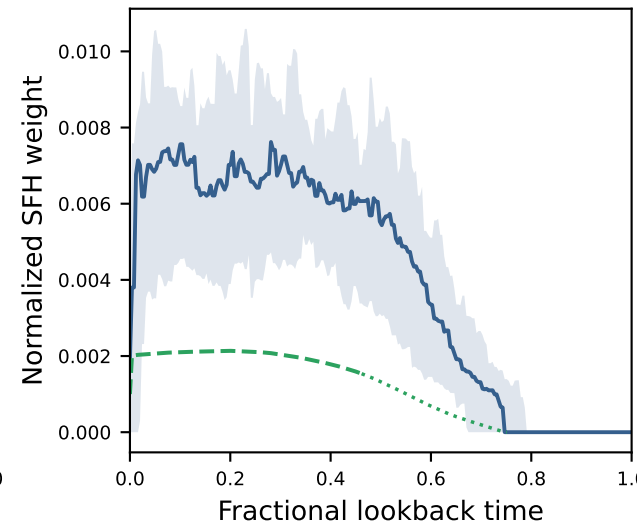
Complete candidate SFH



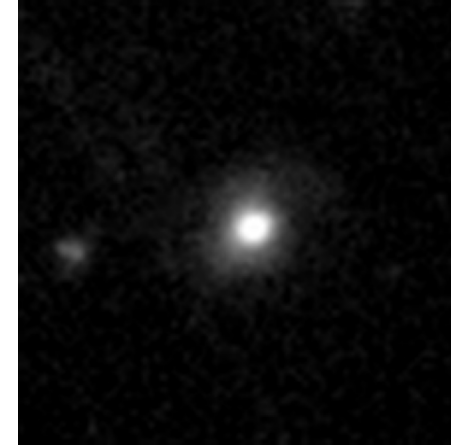
rank 3: 2678711556674304258
 $\log M^*=10.83$ (-0.15) $z=1.334$
 $\Delta MS=+1.77$ $D=0.056$



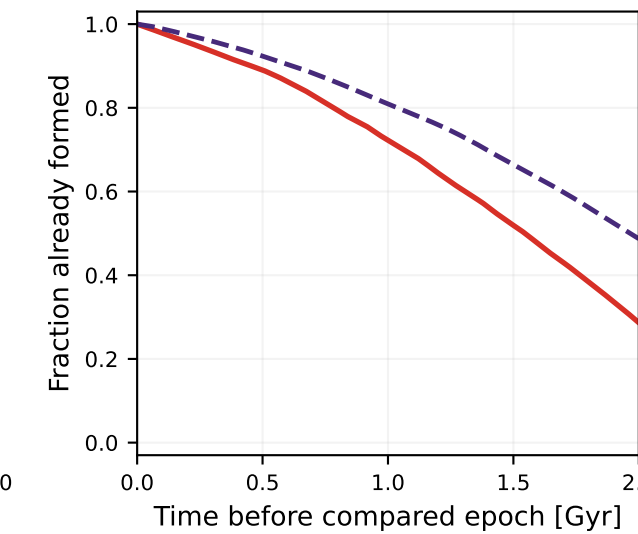
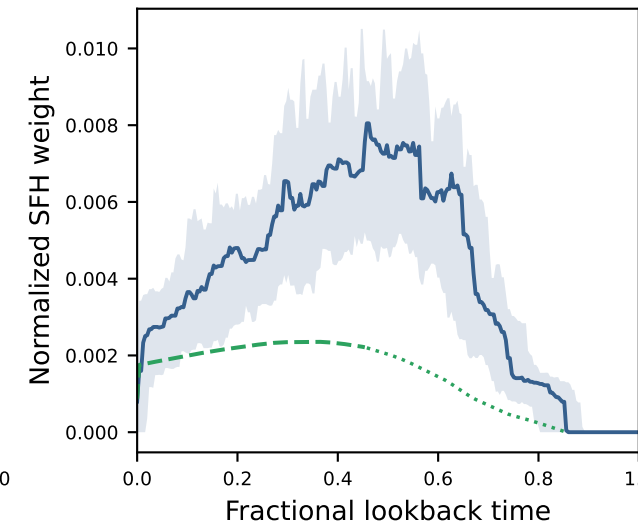
Complete candidate SFH



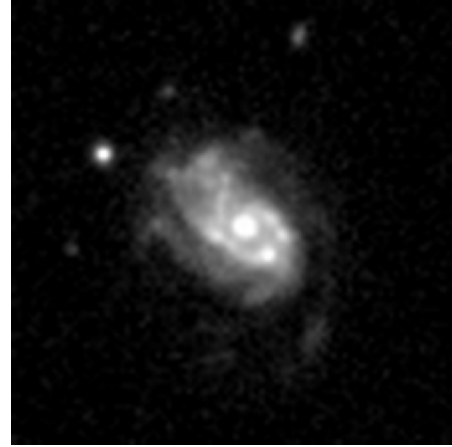
rank 4: 2634119826667500832
 $\log M^*=11.05$ (+0.07) $z=1.336$
 $\Delta MS=+1.56$ $D=0.093$



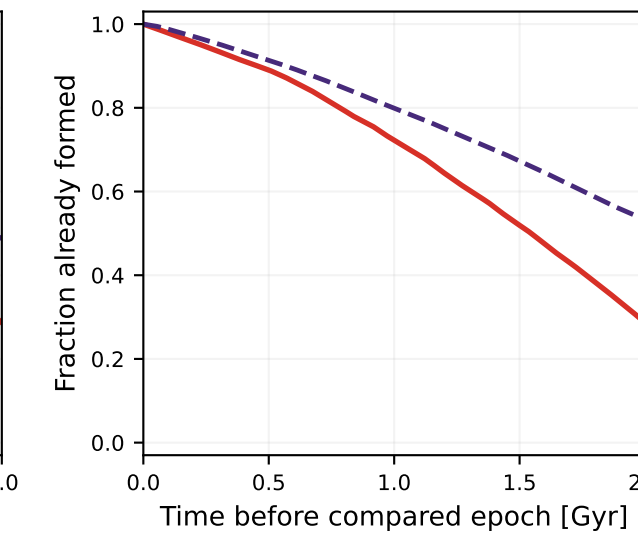
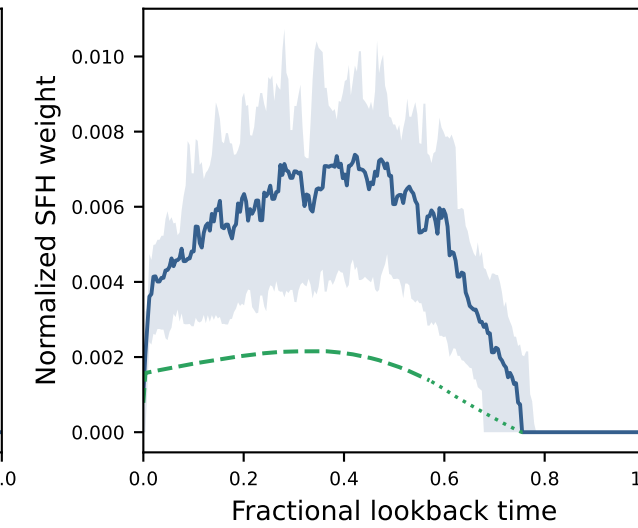
Complete candidate SFH



rank 5: 2675195668637612508
 $\log M^*=10.83$ (-0.15) $z=0.976$
 $\Delta MS=+1.70$ $D=0.093$

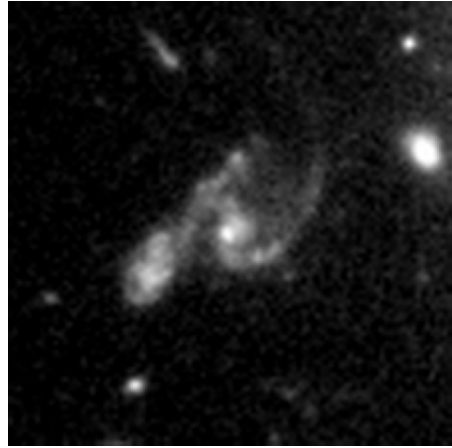


Complete candidate SFH

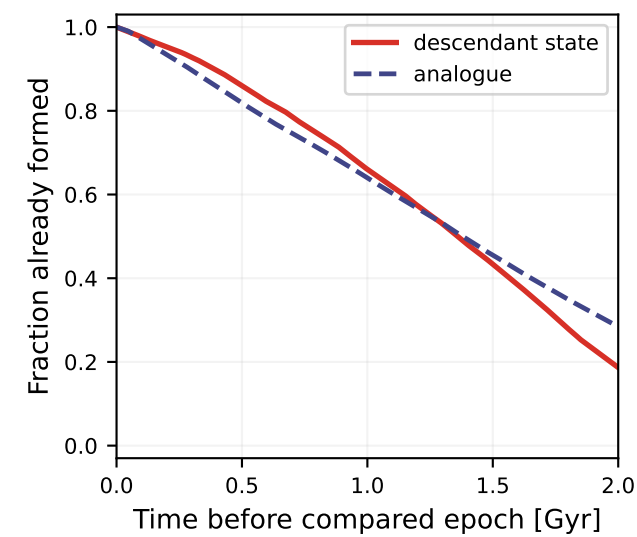
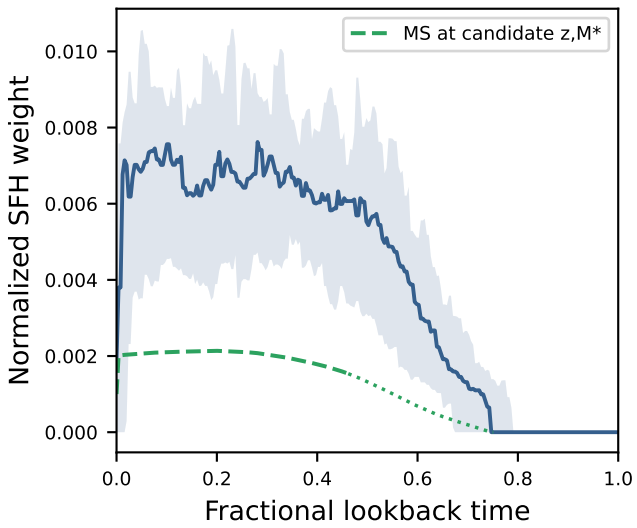


Earlier state $f=0.90$: lookback=3.17 Gyr, predicted $\log M^*=10.95$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

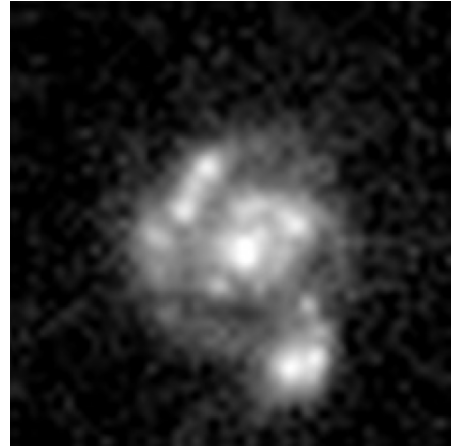
rank 1: 2678711556674304258
 $\log M^*=10.83$ (-0.12) $z=1.334$
 $\Delta MS=+1.77$ $D=0.032$



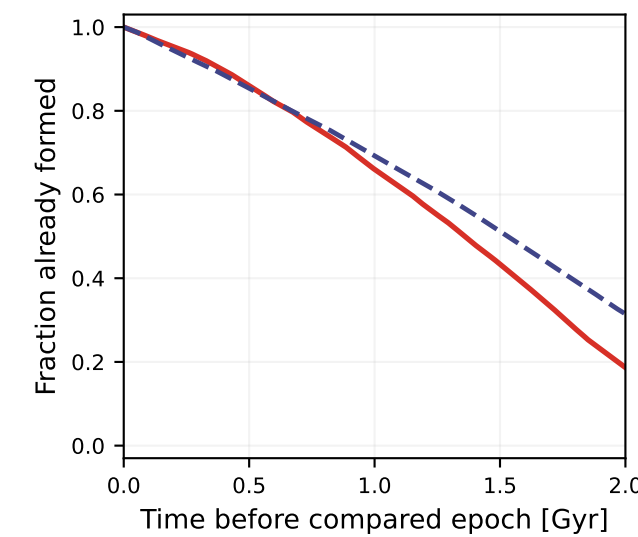
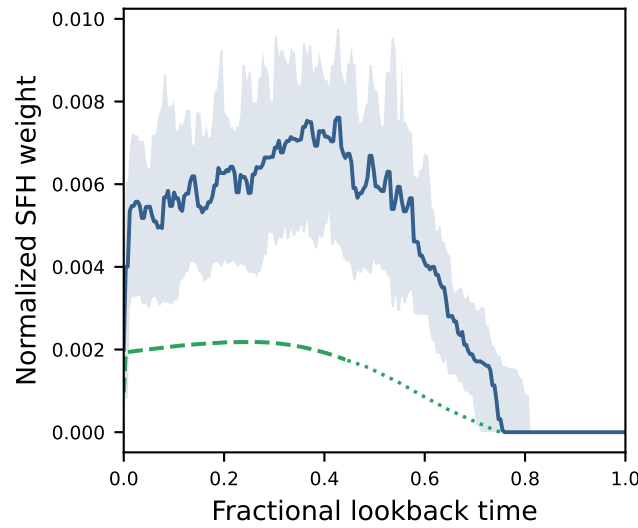
Complete candidate SFH



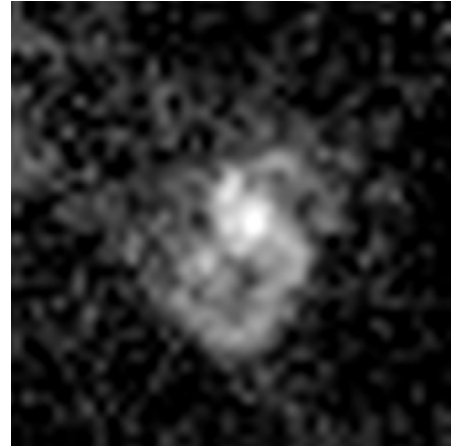
rank 2: 2659854720684798410
 $\log M^*=10.93$ (-0.03) $z=1.378$
 $\Delta MS=+1.77$ $D=0.045$



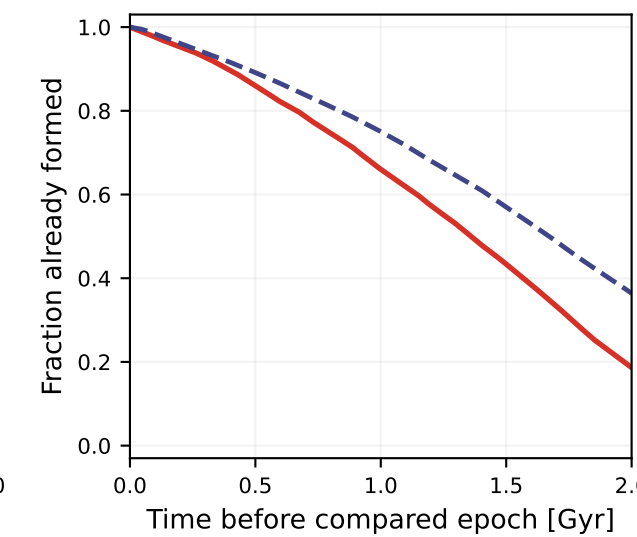
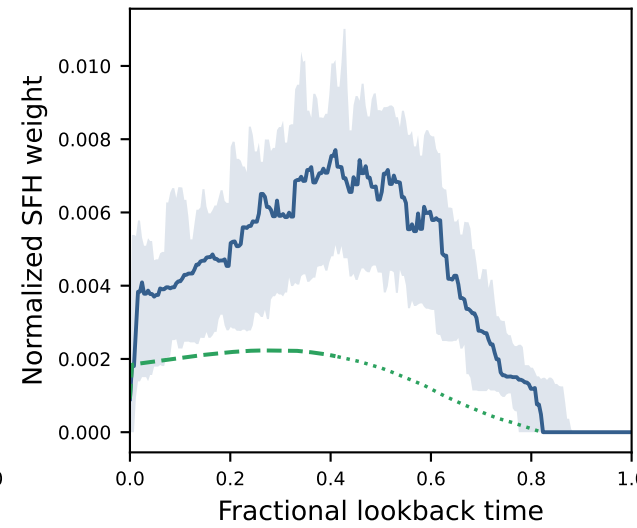
Complete candidate SFH



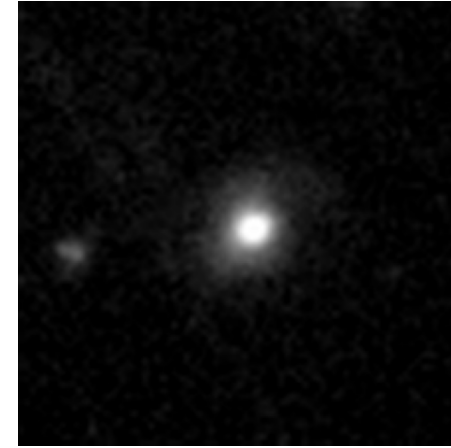
rank 3: 2753316360680891979
 $\log M^*=11.08$ (+0.12) $z=1.472$
 $\Delta MS=+1.77$ $D=0.086$



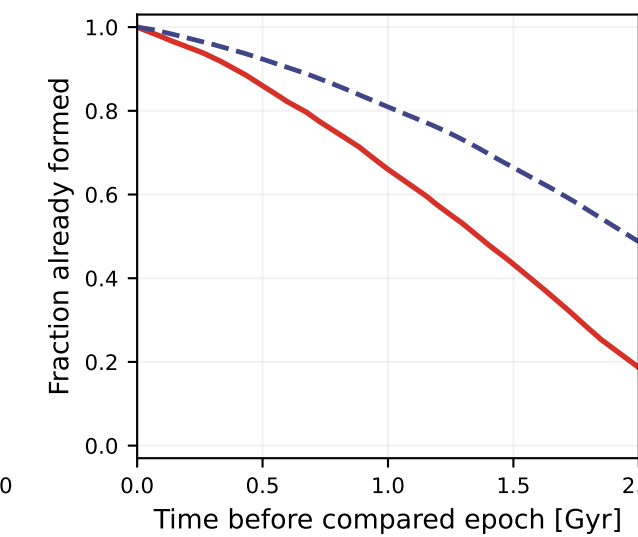
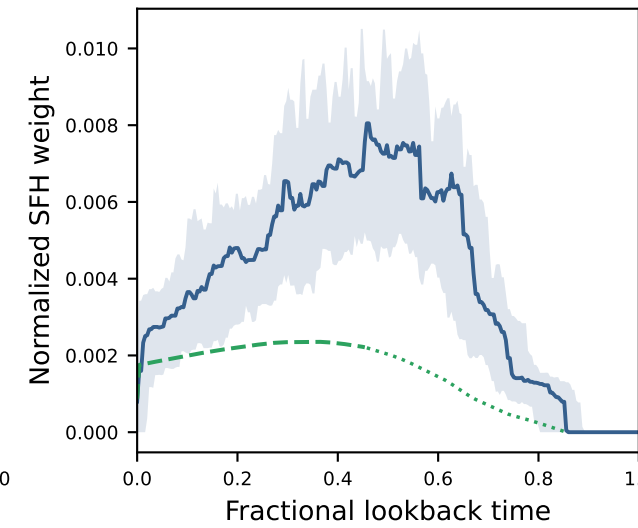
Complete candidate SFH



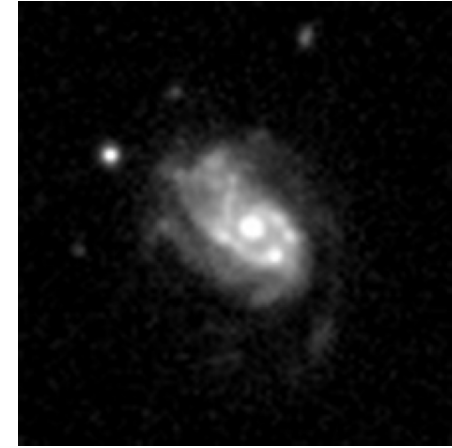
rank 4: 2634119826667500832
 $\log M^*=11.05$ (+0.10) $z=1.336$
 $\Delta MS=+1.56$ $D=0.149$



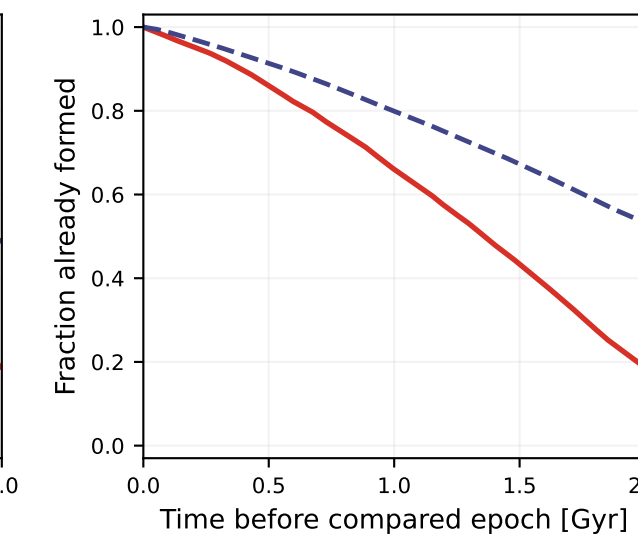
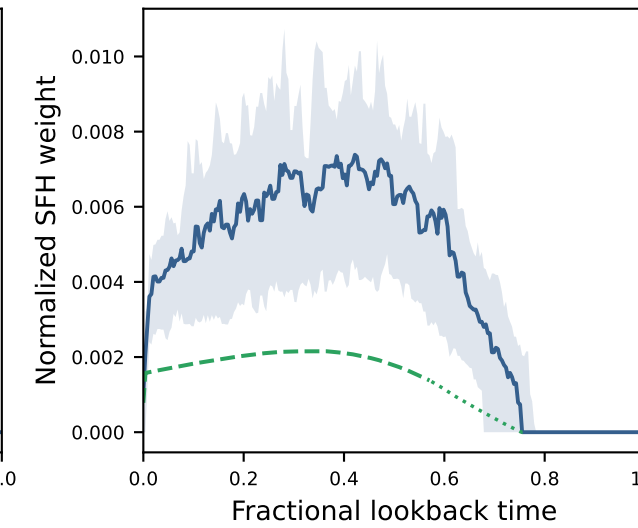
Complete candidate SFH



rank 5: 2675195668637612508
 $\log M^*=10.83$ (-0.12) $z=0.976$
 $\Delta MS=+1.70$ $D=0.150$

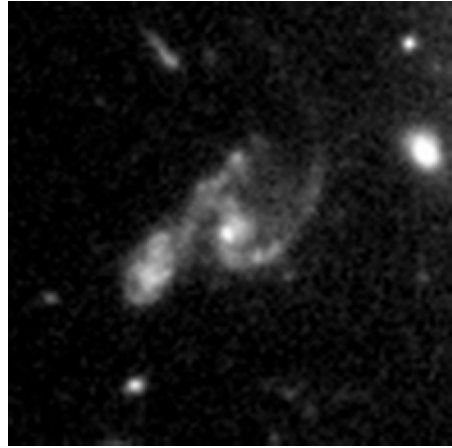


Complete candidate SFH

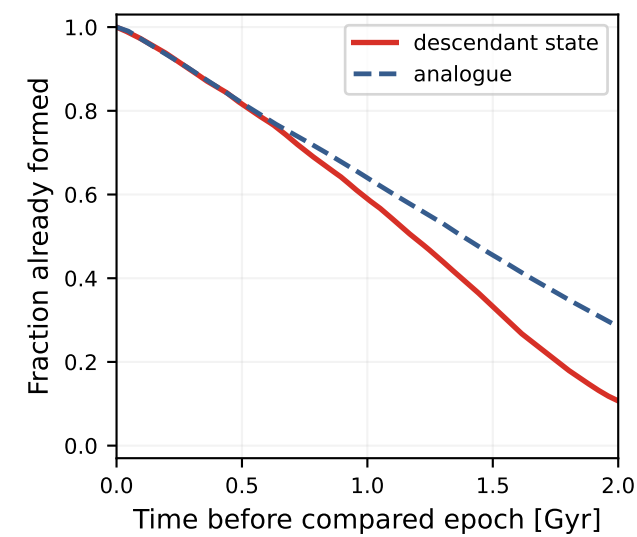
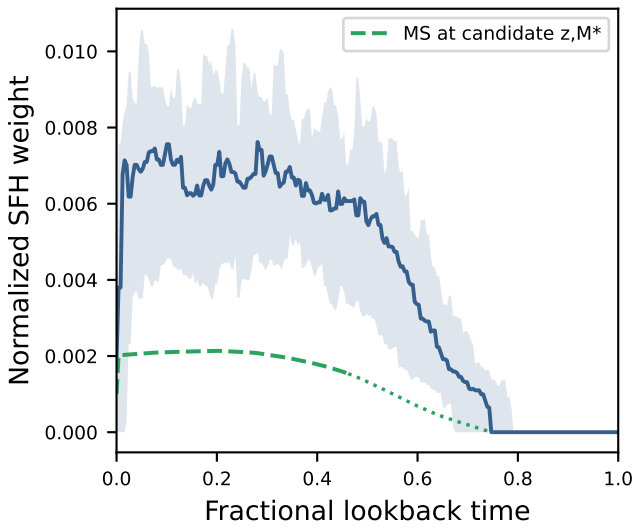


Earlier state $f=0.85$: lookback=3.41 Gyr, predicted $\log M^*=10.93$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

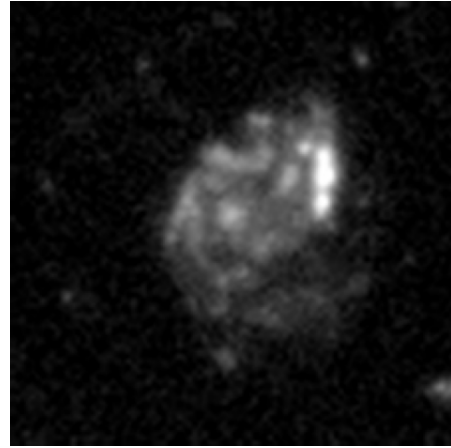
rank 1: 2678711556674304258
 $\log M^*=10.83$ (-0.10) $z=1.334$
 $\Delta MS=+1.77$ $D=0.067$



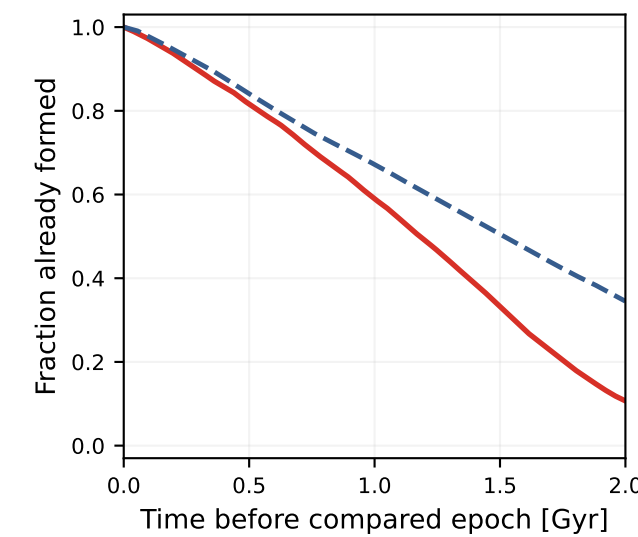
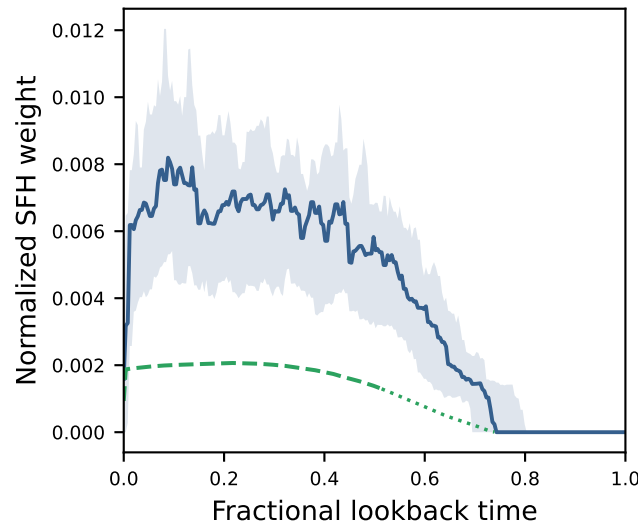
Complete candidate SFH



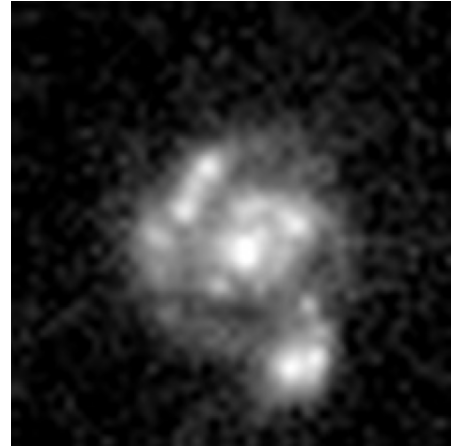
rank 2: 2703488790670980759
 $\log M^*=10.80$ (-0.13) $z=1.170$
 $\Delta MS=+1.64$ $D=0.100$



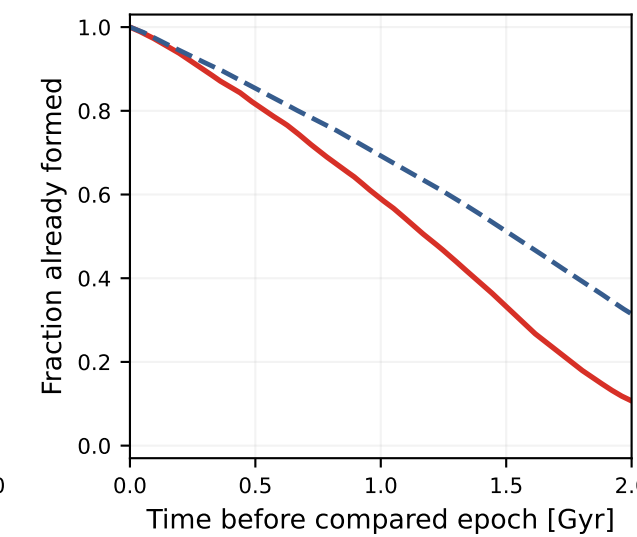
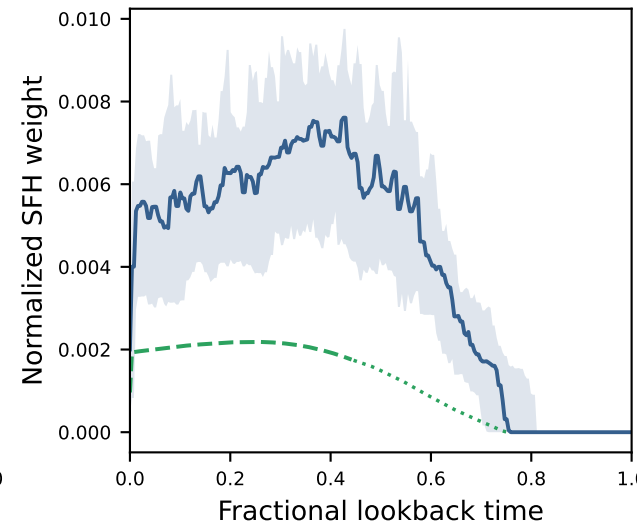
Complete candidate SFH



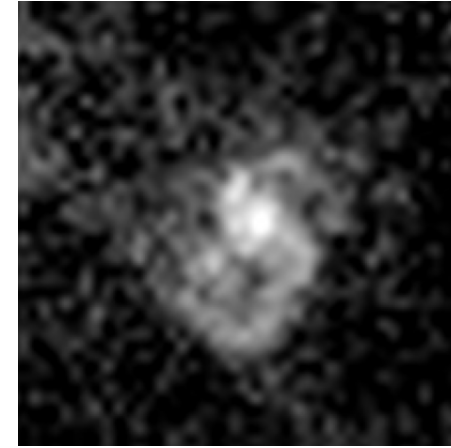
rank 3: 2659854720684798410
 $\log M^*=10.93$ (-0.00) $z=1.378$
 $\Delta MS=+1.77$ $D=0.108$



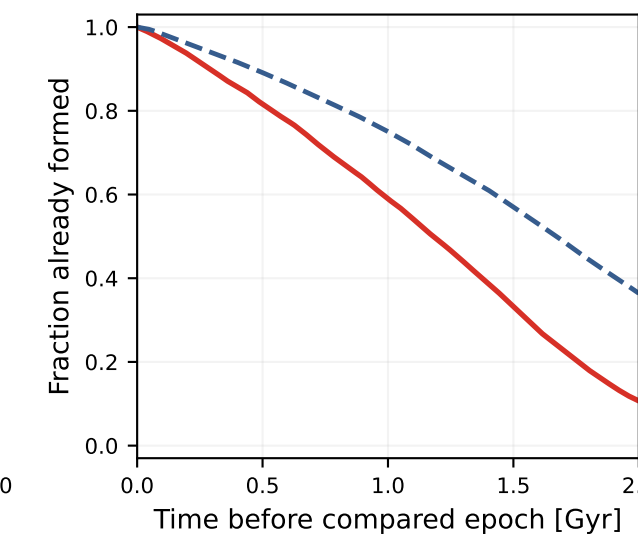
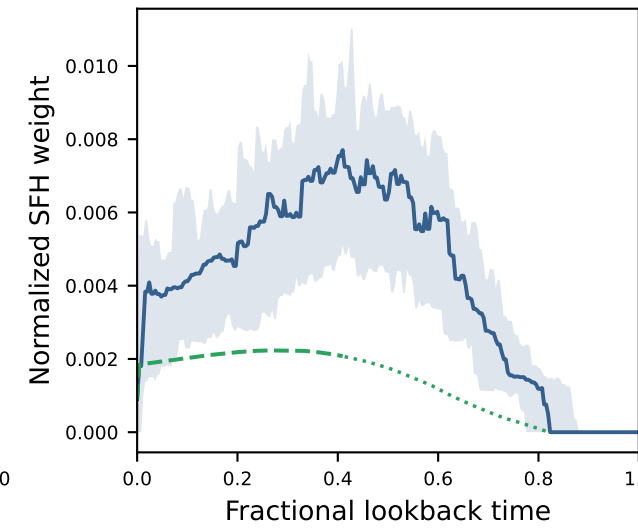
Complete candidate SFH



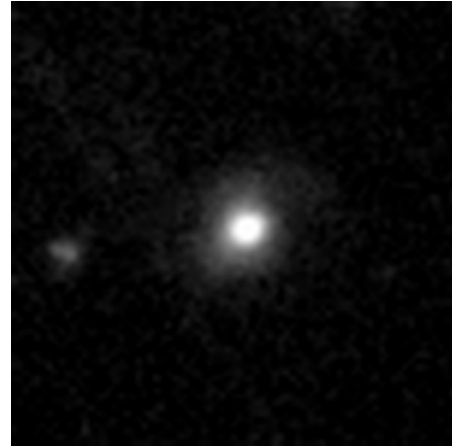
rank 4: 2753316360680891979
 $\log M^*=11.08$ (+0.15) $z=1.472$
 $\Delta MS=+1.77$ $D=0.152$



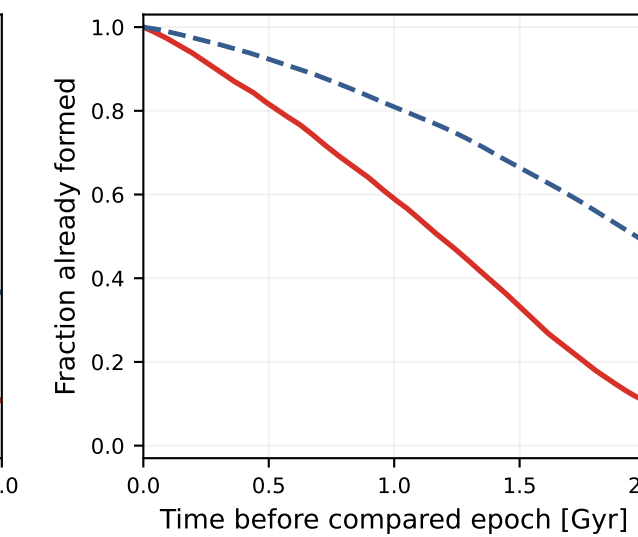
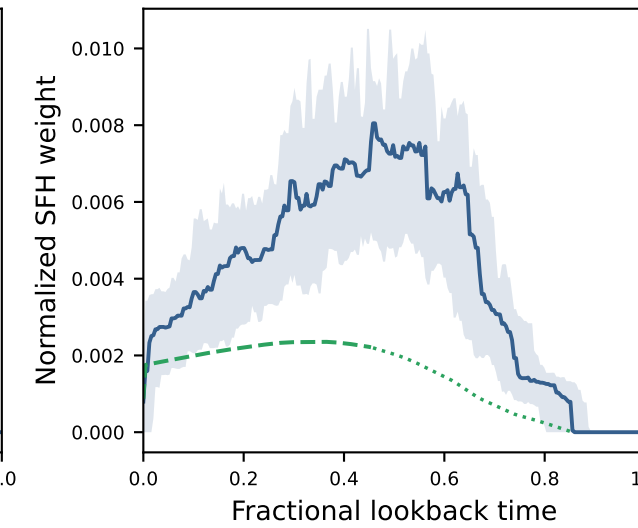
Complete candidate SFH



rank 5: 2634119826667500832
 $\log M^*=11.05$ (+0.12) $z=1.336$
 $\Delta MS=+1.56$ $D=0.216$

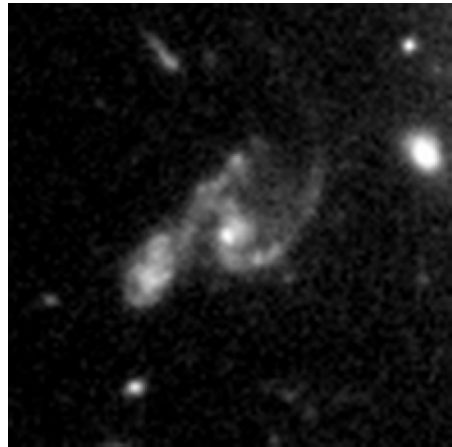


Complete candidate SFH

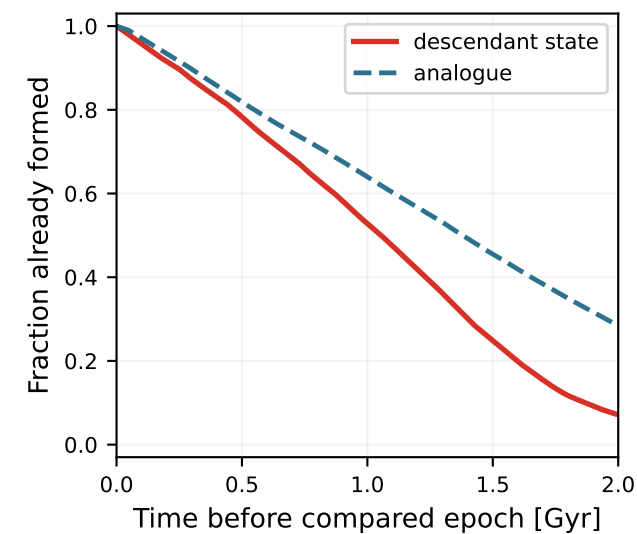
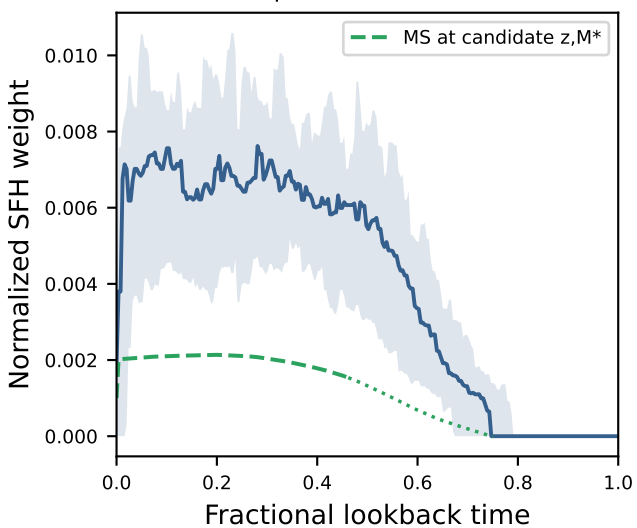


Earlier state $f=0.80$: lookback=3.60 Gyr, predicted $\log M^*=10.90$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

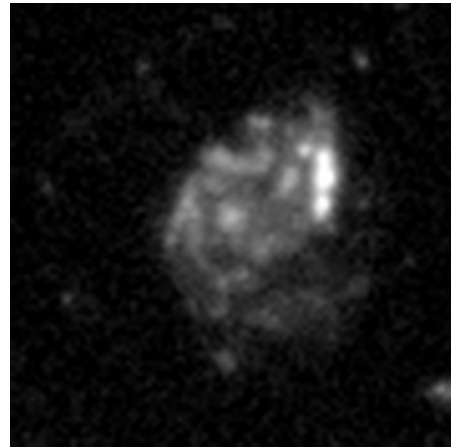
rank 1: 2678711556674304258
 $\log M^*=10.83$ (-0.07) $z=1.334$
 $\Delta MS=+1.77$ $D=0.119$



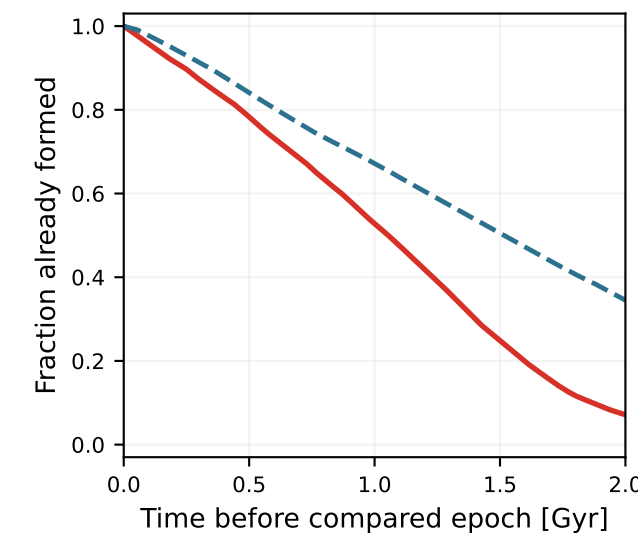
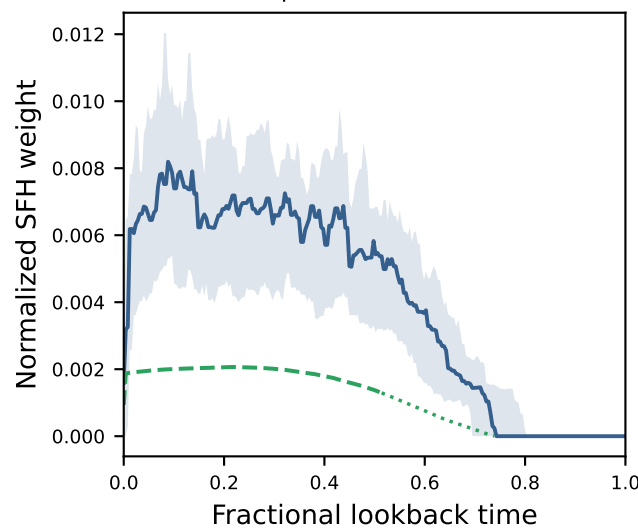
Complete candidate SFH



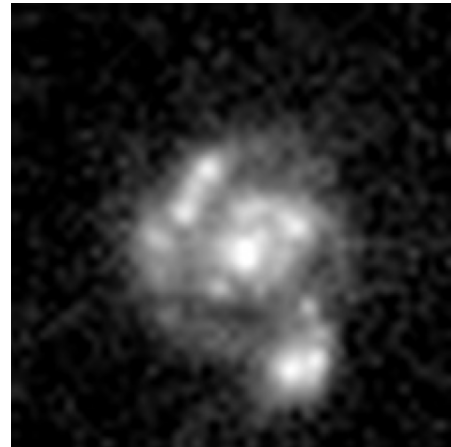
rank 2: 2703488790670980759
 $\log M^*=10.80$ (-0.11) $z=1.170$
 $\Delta MS=+1.64$ $D=0.152$



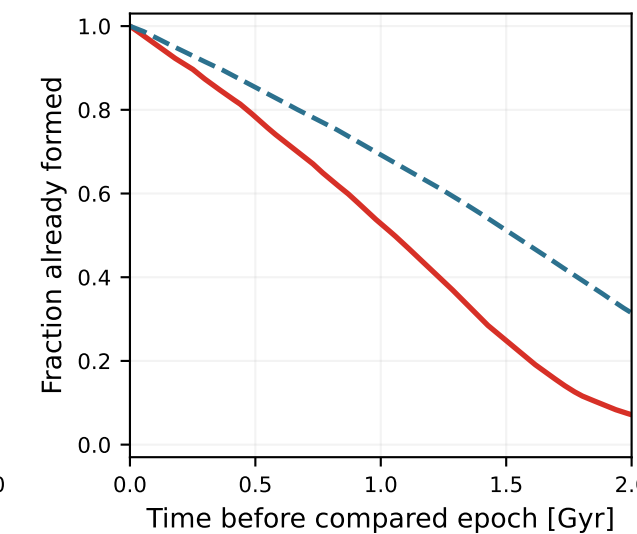
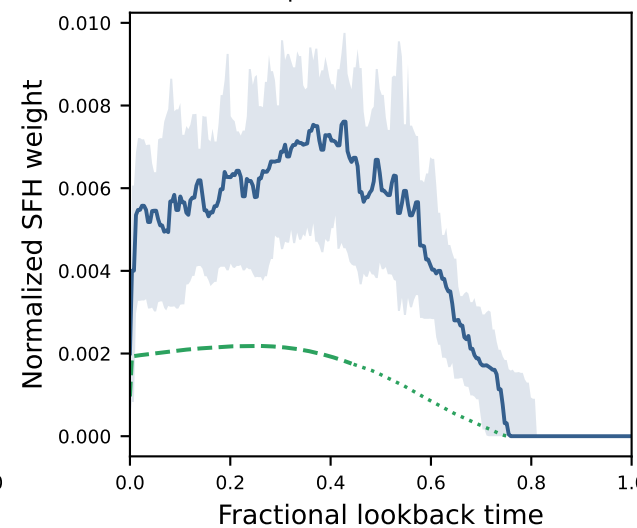
Complete candidate SFH



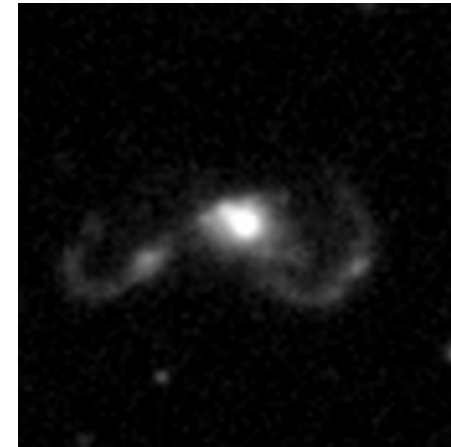
rank 3: 2659854720684798410
 $\log M^*=10.93$ (+0.03) $z=1.378$
 $\Delta MS=+1.77$ $D=0.160$



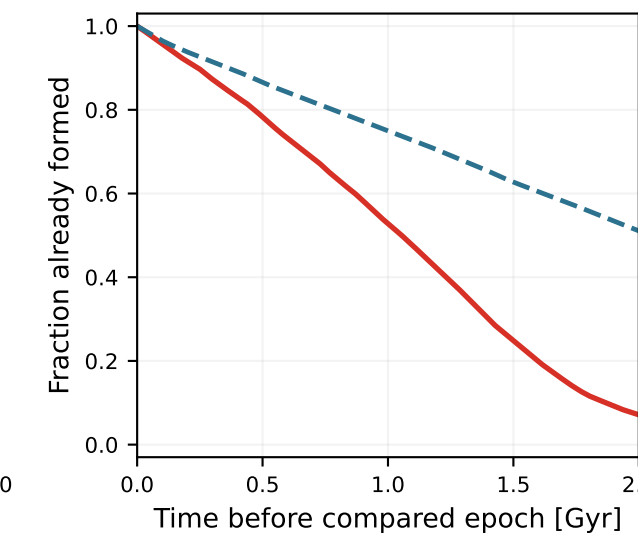
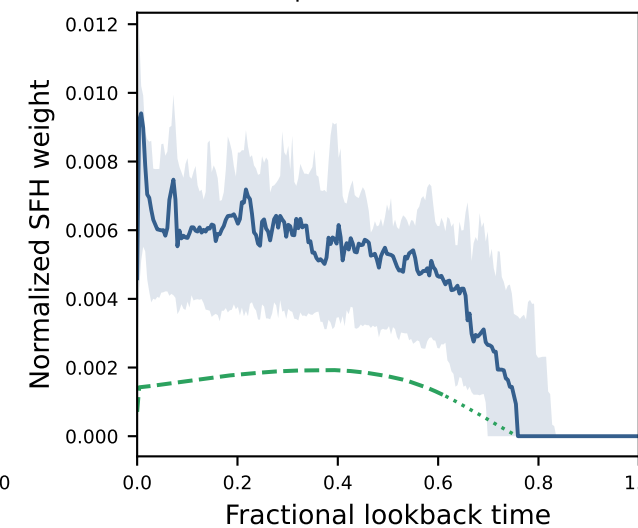
Complete candidate SFH



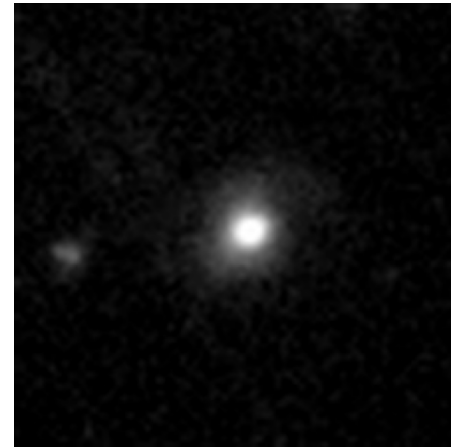
rank 4: 2649173403675031662
 $\log M^*=10.77$ (-0.14) $z=0.832$
 $\Delta MS=+0.66$ $D=0.228$



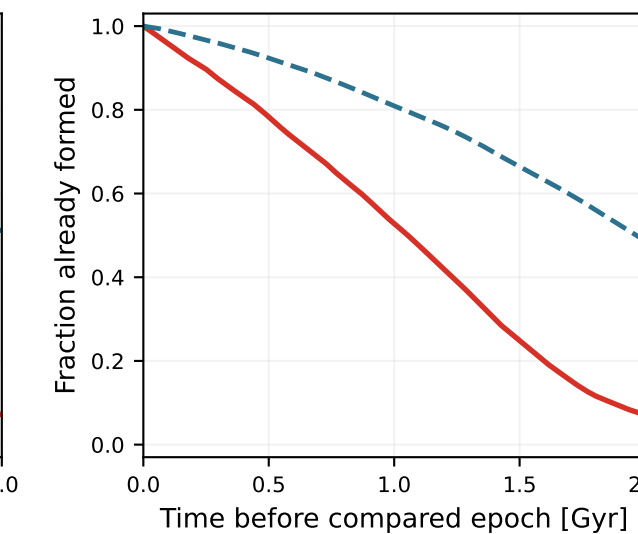
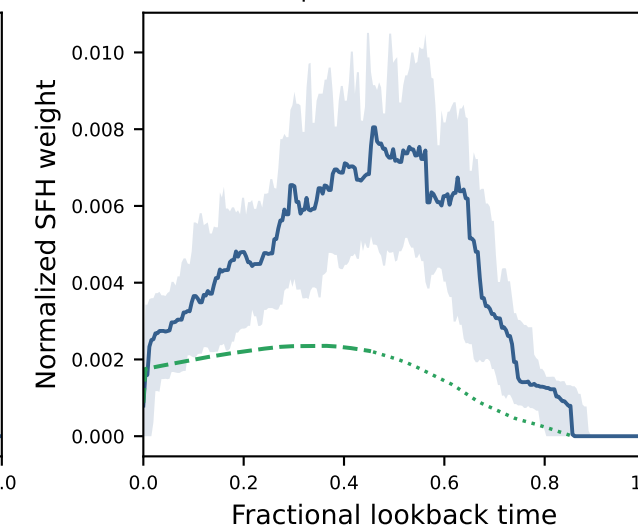
Complete candidate SFH



rank 5: 2634119826667500832
 $\log M^*=11.05$ (+0.15) $z=1.336$
 $\Delta MS=+1.56$ $D=0.268$

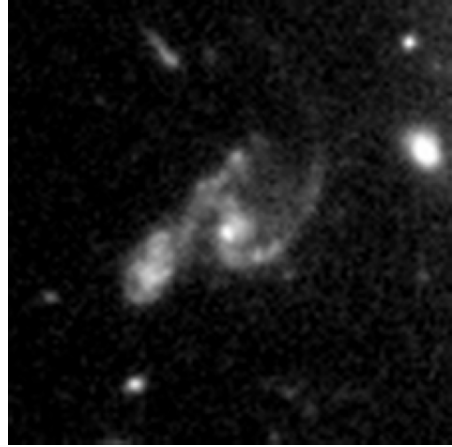


Complete candidate SFH

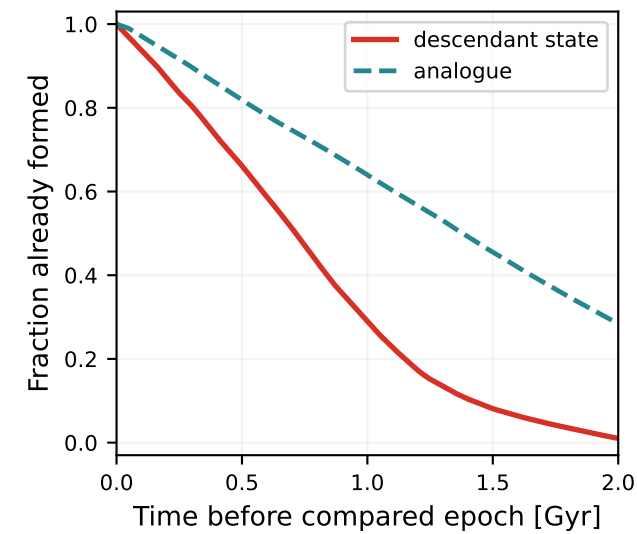
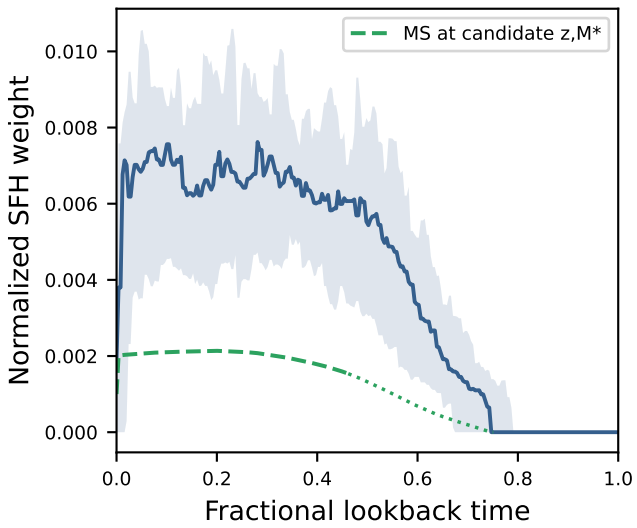


Earlier state $f=0.60$: lookback=4.16 Gyr, predicted $\log M^*=10.78$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

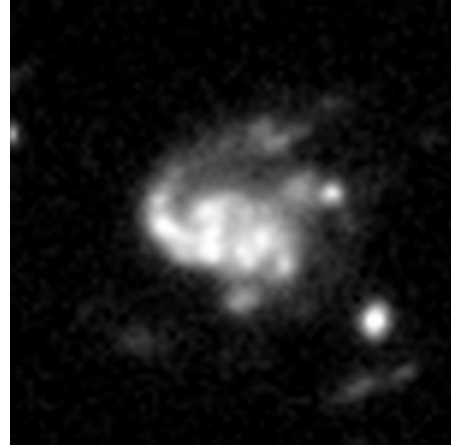
rank 1: 2678711556674304258
 $\log M^*=10.83 (+0.05)$ $z=1.334$
 $\Delta MS=+1.77$ $D=0.261$



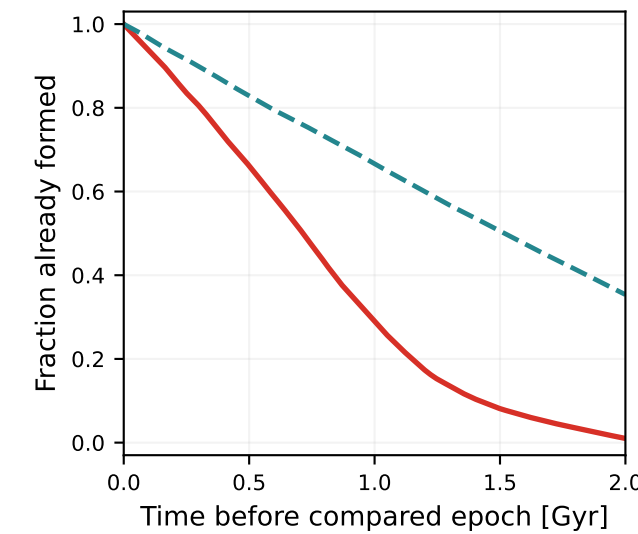
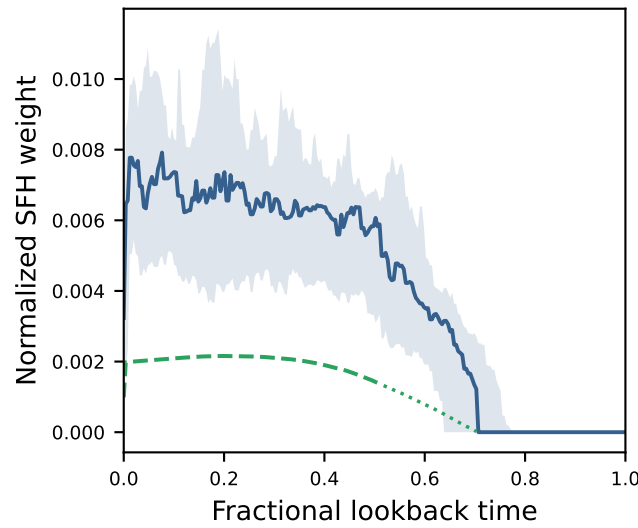
Complete candidate SFH



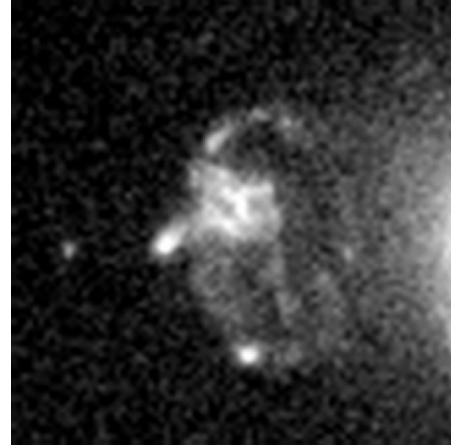
rank 2: 2741447642657085578
 $\log M^*=10.70 (-0.08)$ $z=1.154$
 $\Delta MS=+1.74$ $D=0.290$



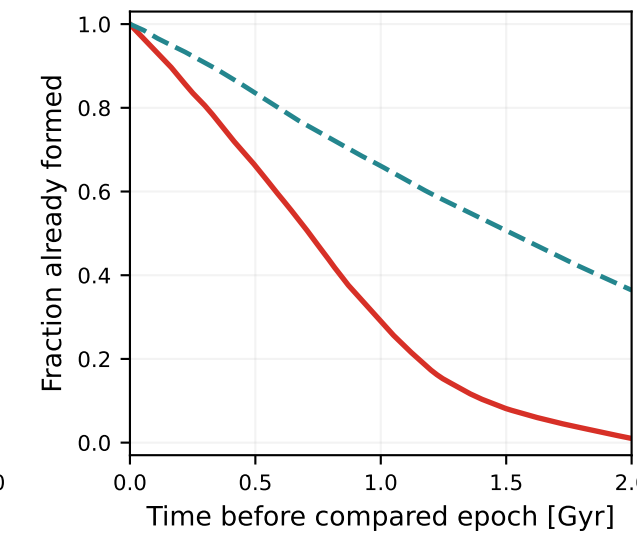
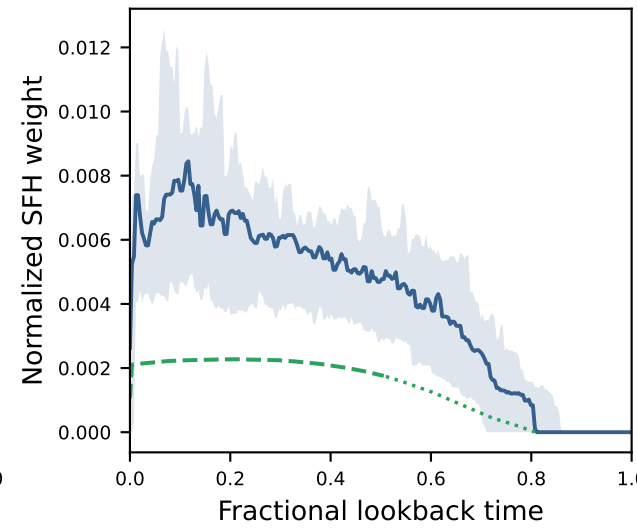
Complete candidate SFH



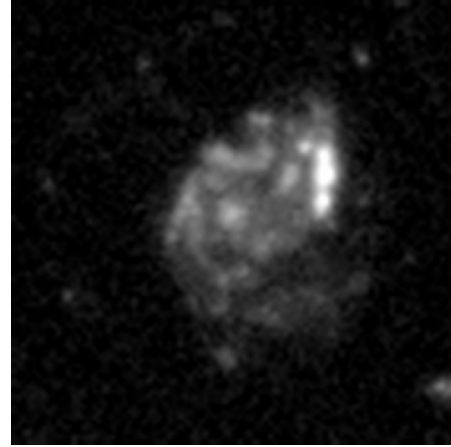
rank 3: 2629425863659052113
 $\log M^*=10.63 (-0.15)$ $z=1.184$
 $\Delta MS=+1.69$ $D=0.291$



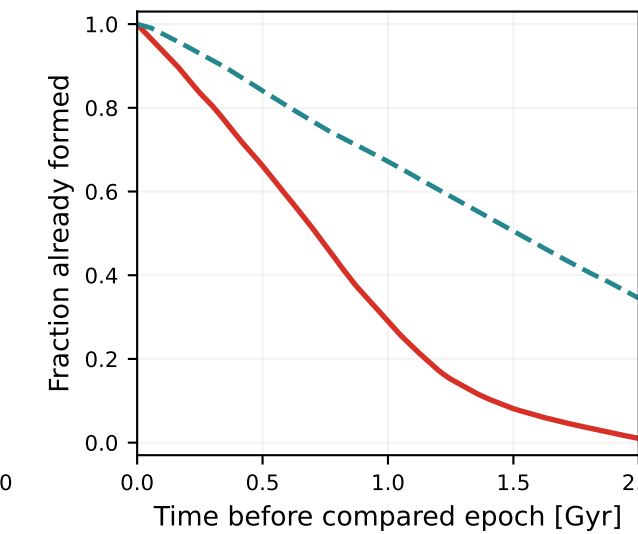
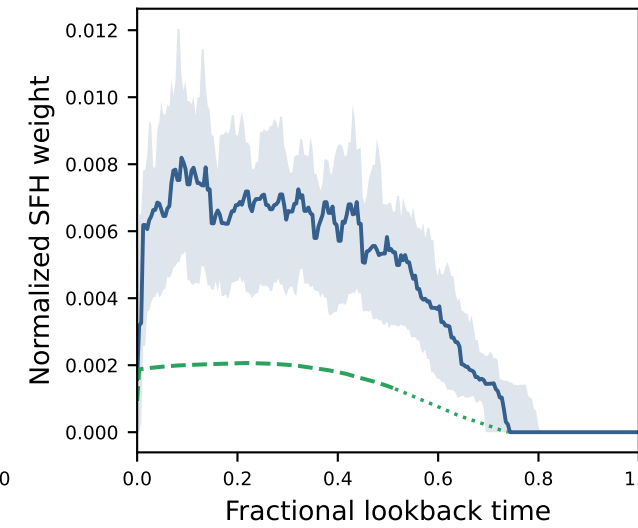
Complete candidate SFH



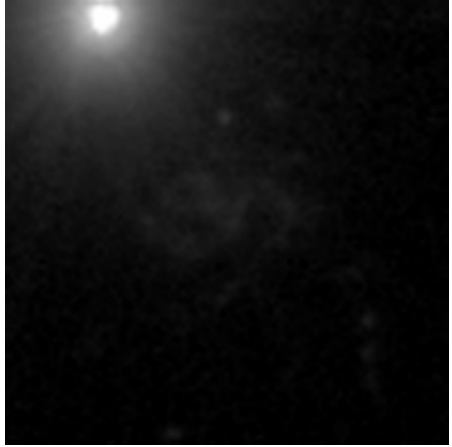
rank 4: 2703488790670980759
 $\log M^*=10.80 (+0.02)$ $z=1.170$
 $\Delta MS=+1.64$ $D=0.294$



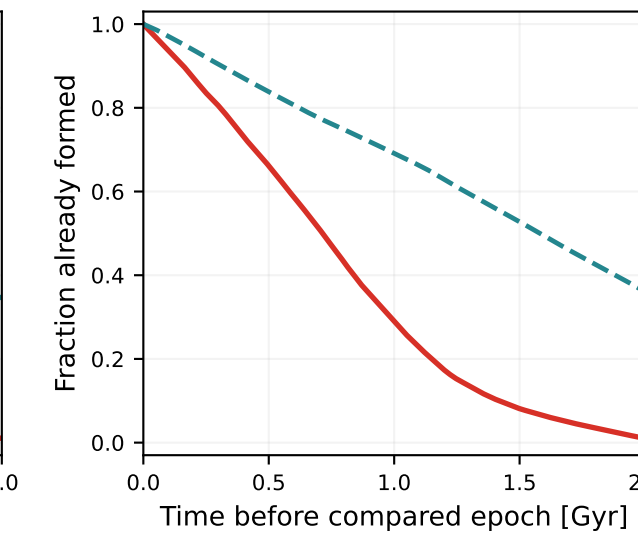
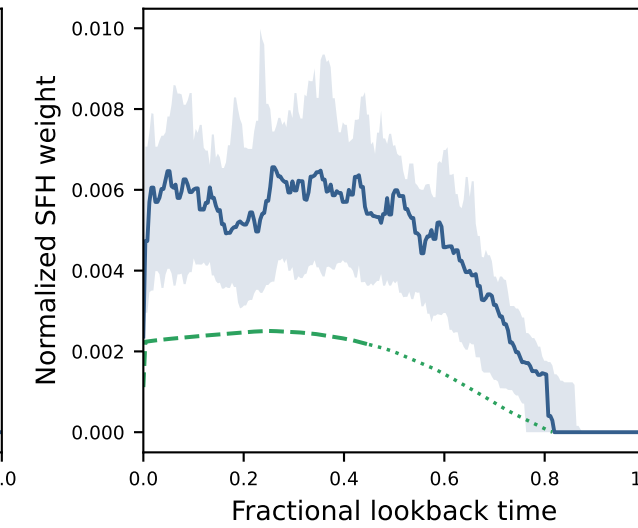
Complete candidate SFH



rank 5: 2652718793642424678
 $\log M^*=10.69 (-0.09)$ $z=1.370$
 $\Delta MS=+1.85$ $D=0.306$

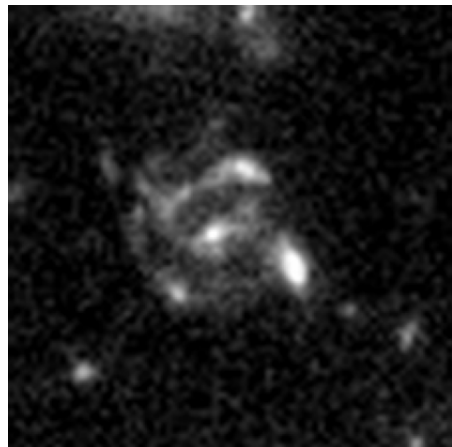


Complete candidate SFH

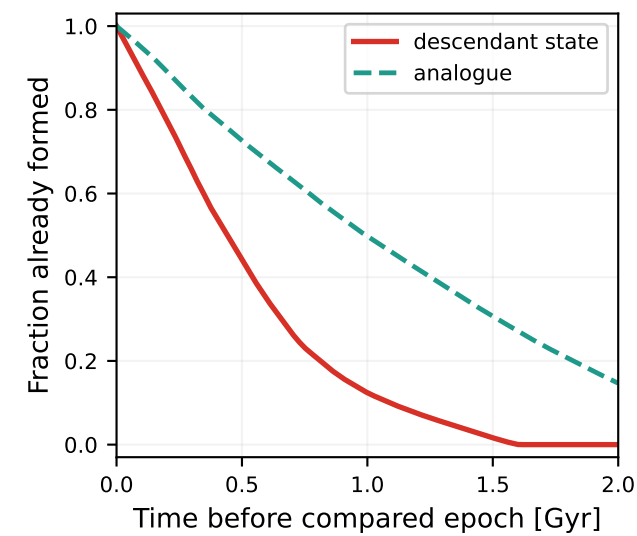
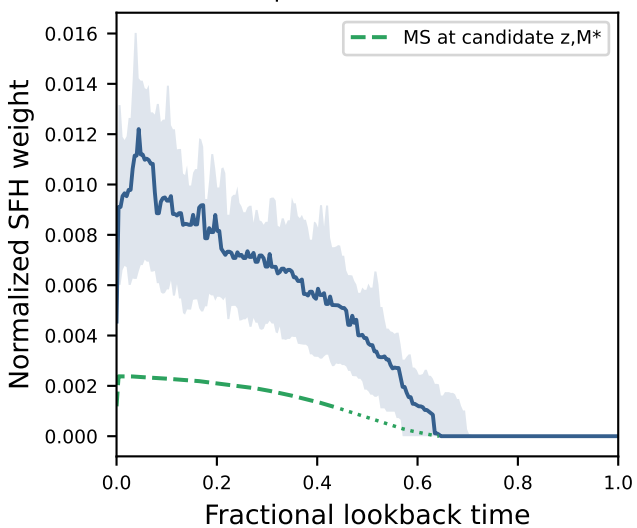


Earlier state $f=0.40$: lookback=4.65 Gyr, predicted $\log M^*=10.60$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

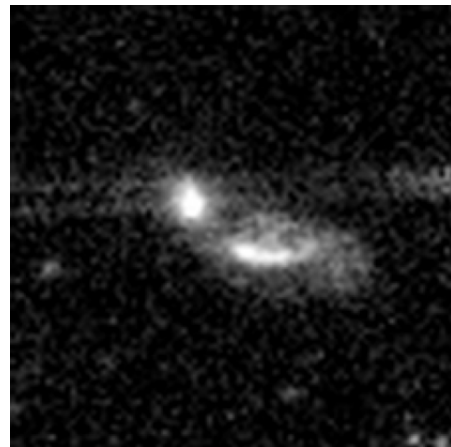
rank 1: 2716117201644464606
 $\log M^*=10.58$ (-0.02) $z=1.350$
 $\Delta MS=+1.46$ $D=0.265$



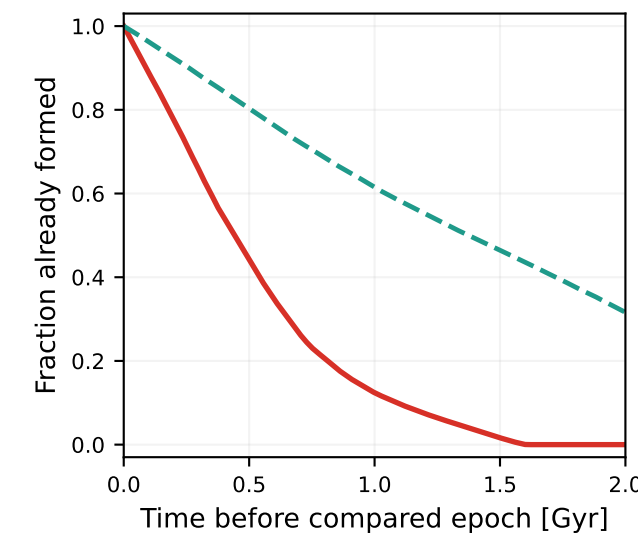
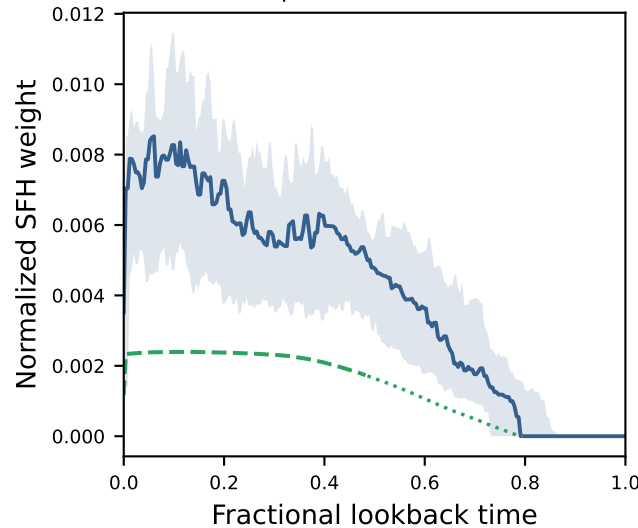
Complete candidate SFH



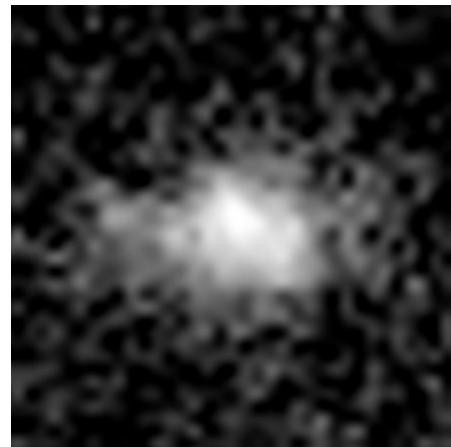
rank 2: 2754251518642661053
 $\log M^*=10.53$ (-0.07) $z=1.250$
 $\Delta MS=+1.85$ $D=0.375$



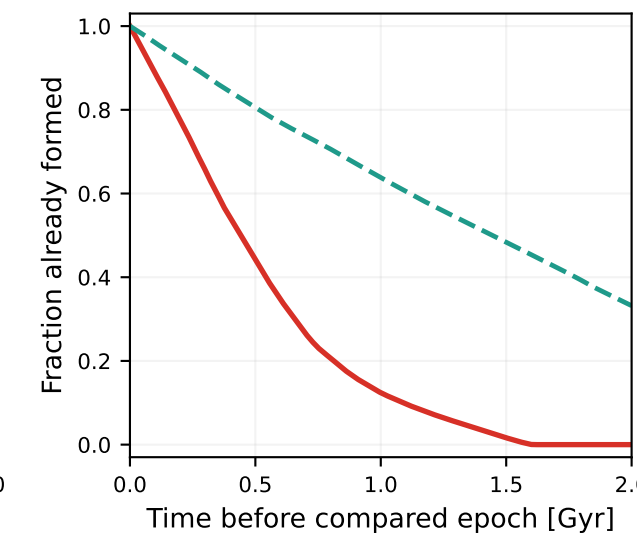
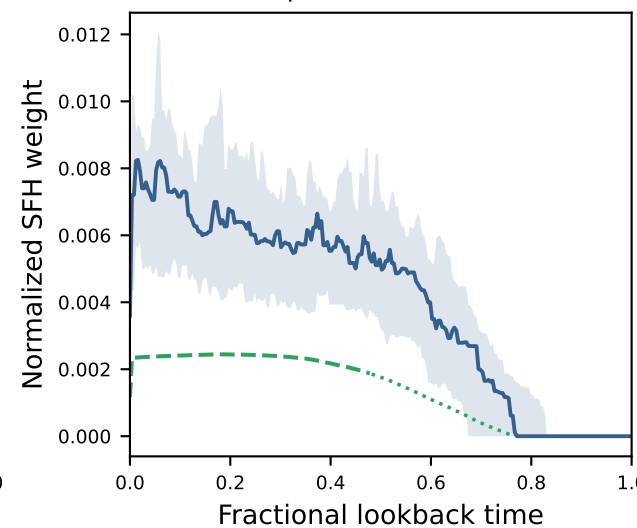
Complete candidate SFH



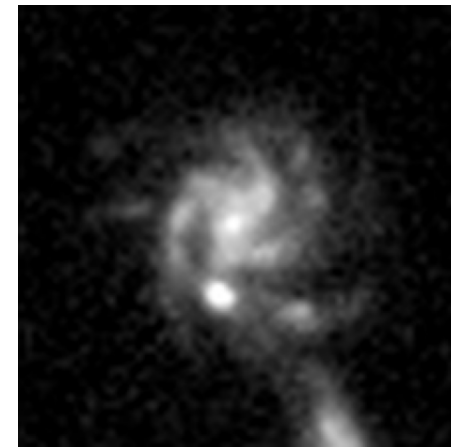
rank 3: 2748216818639415066
 $\log M^*=10.54$ (-0.06) $z=1.266$
 $\Delta MS=+1.89$ $D=0.388$



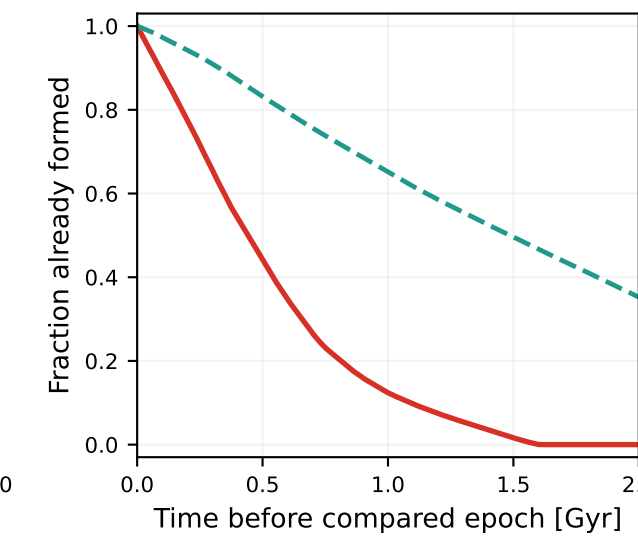
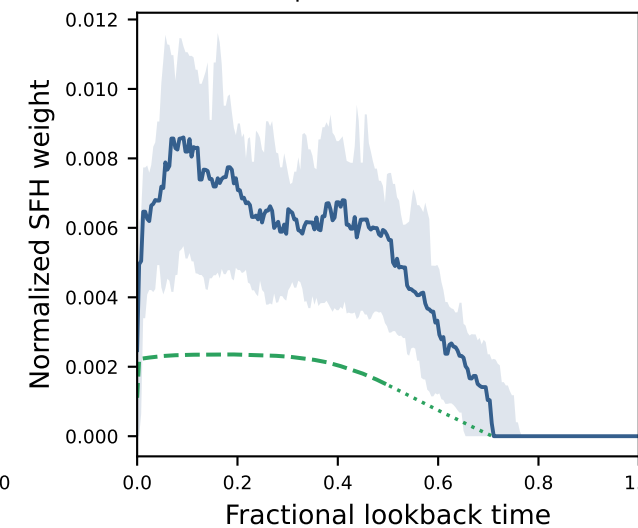
Complete candidate SFH



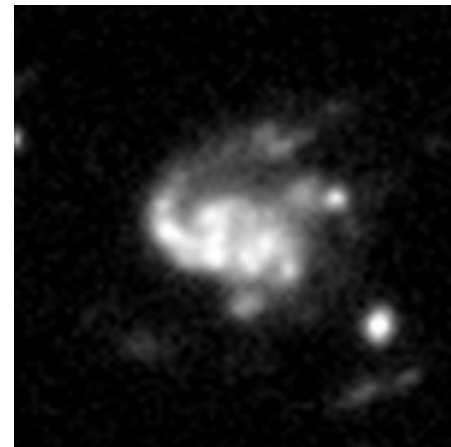
rank 4: 2674024524656252959
 $\log M^*=10.49$ (-0.11) $z=1.118$
 $\Delta MS=+1.43$ $D=0.405$



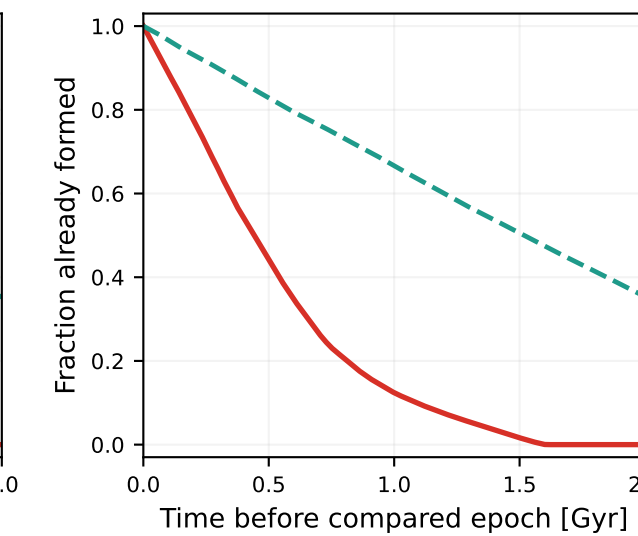
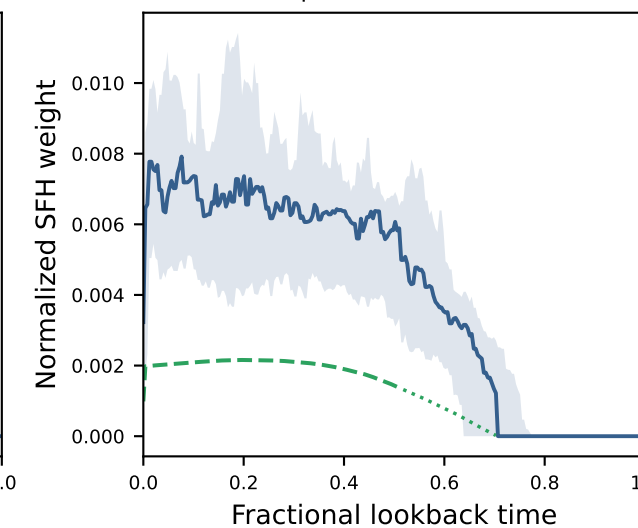
Complete candidate SFH



rank 5: 2741447642657085578
 $\log M^*=10.70$ (+0.10) $z=1.154$
 $\Delta MS=+1.74$ $D=0.409$

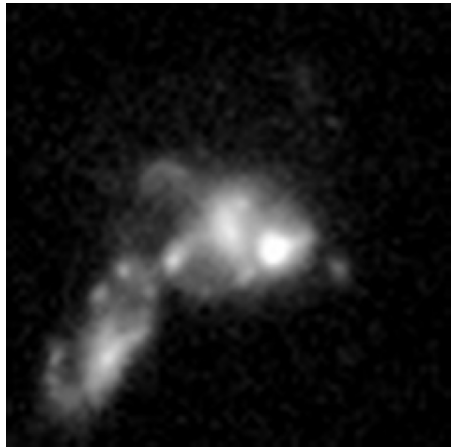


Complete candidate SFH

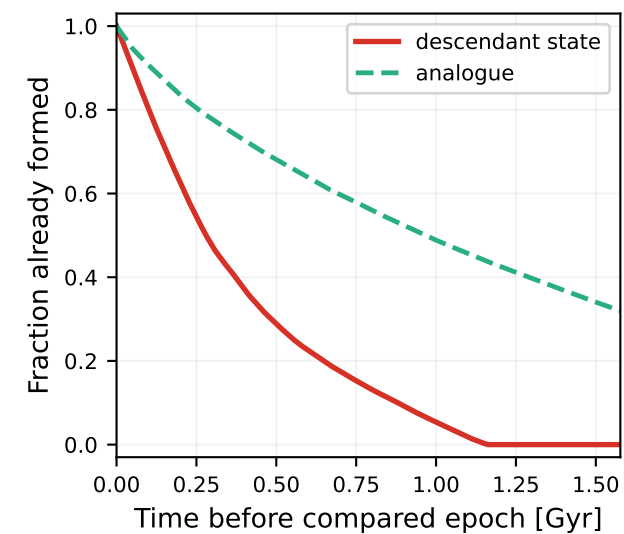
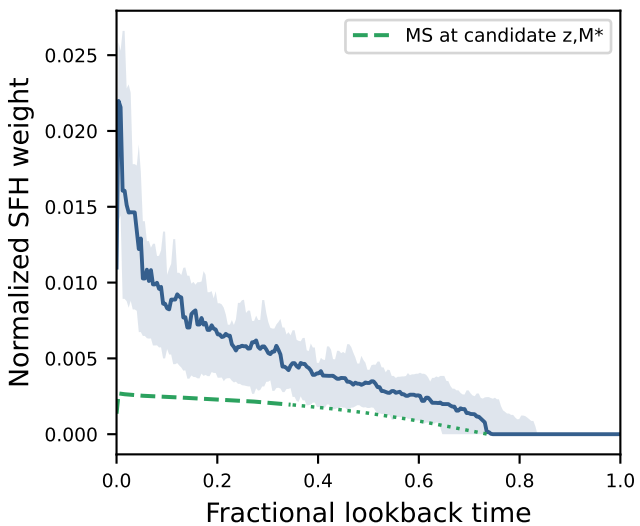


Earlier state $f=0.20$: lookback=5.09 Gyr, predicted $\log M^*=10.30$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

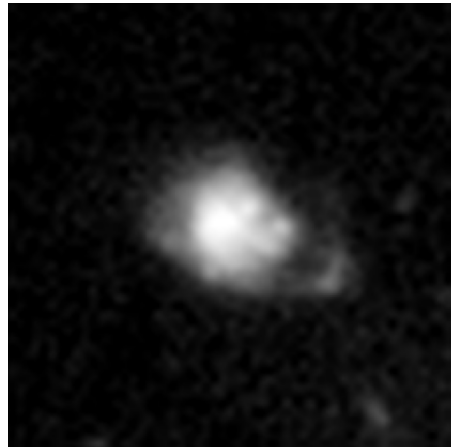
rank 1: 2741023943678283074
 $\log M^*=10.26$ (-0.04) $z=1.194$
 $\Delta MS=+1.94$ $D=0.352$



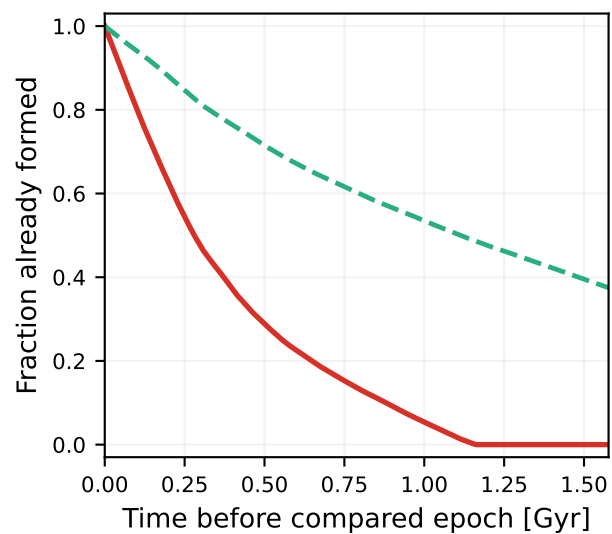
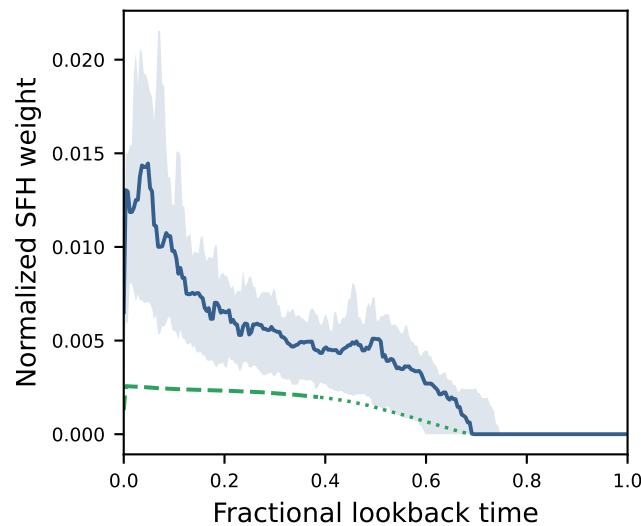
Complete candidate SFH



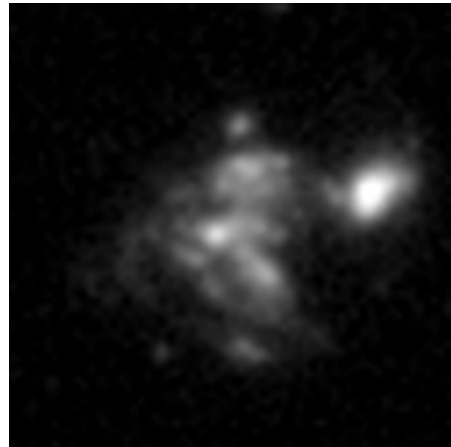
rank 2: 2656685702653947484
 $\log M^*=10.28$ (-0.02) $z=1.122$
 $\Delta MS=+1.26$ $D=0.394$



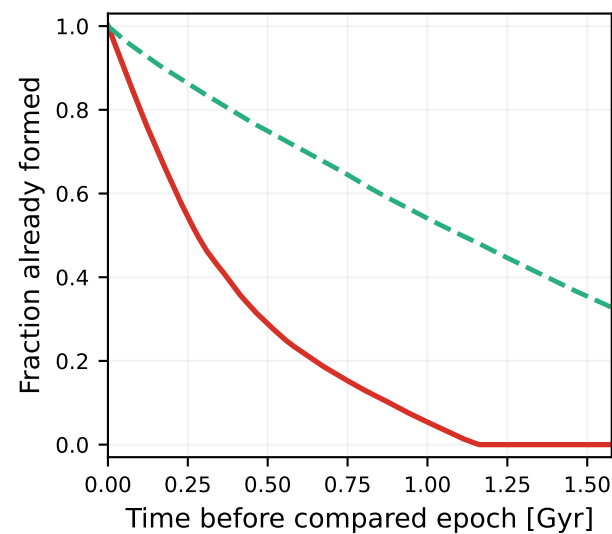
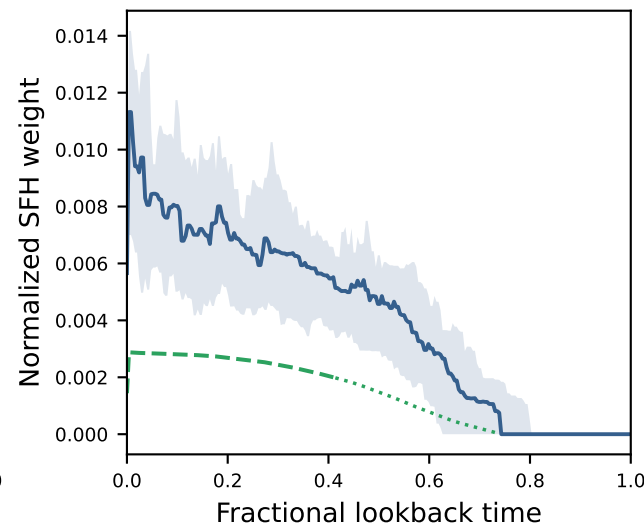
Complete candidate SFH



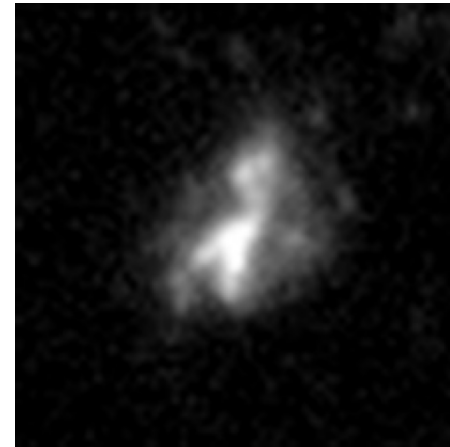
rank 3: 2640109785664925808
 $\log M^*=10.35$ (+0.05) $z=1.478$
 $\Delta MS=+1.75$ $D=0.399$



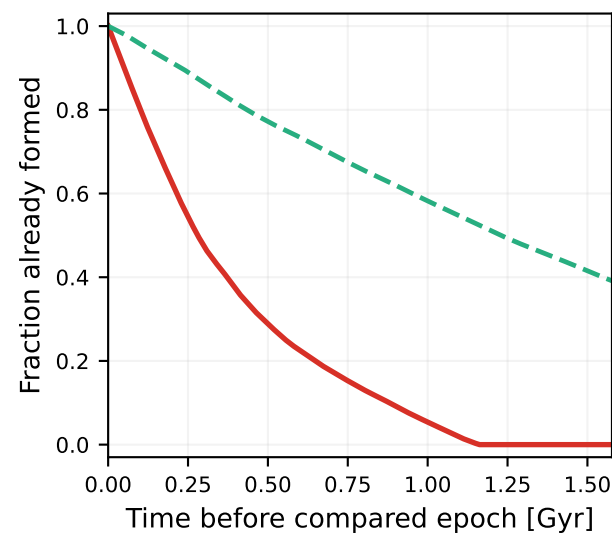
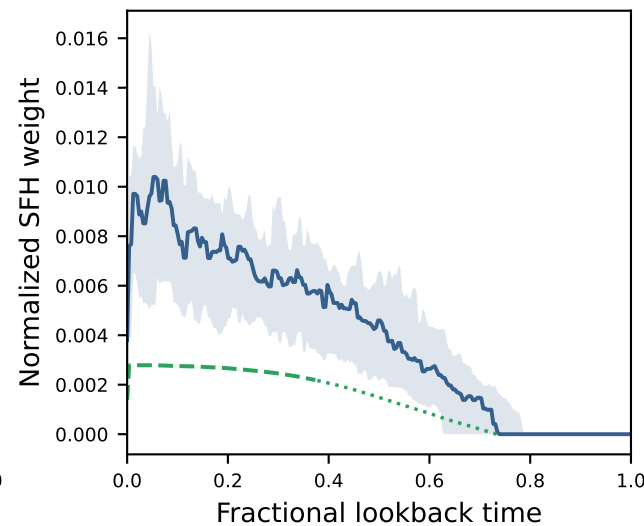
Complete candidate SFH



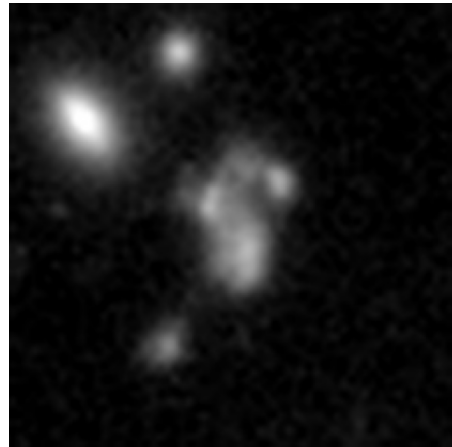
rank 4: 2717353722653902287
 $\log M^*=10.23$ (-0.07) $z=1.198$
 $\Delta MS=+1.81$ $D=0.435$



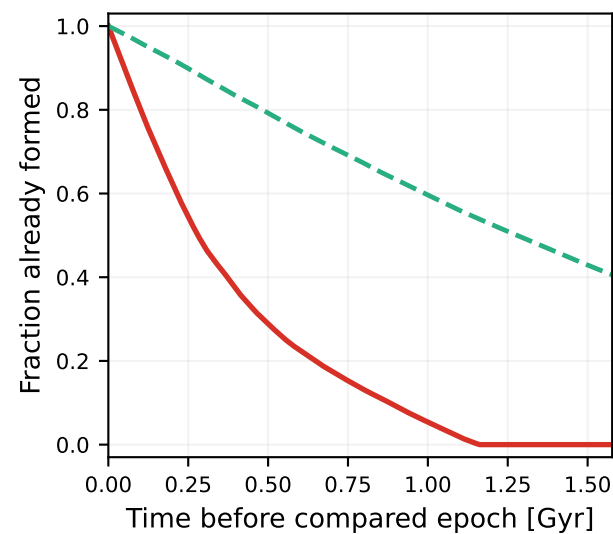
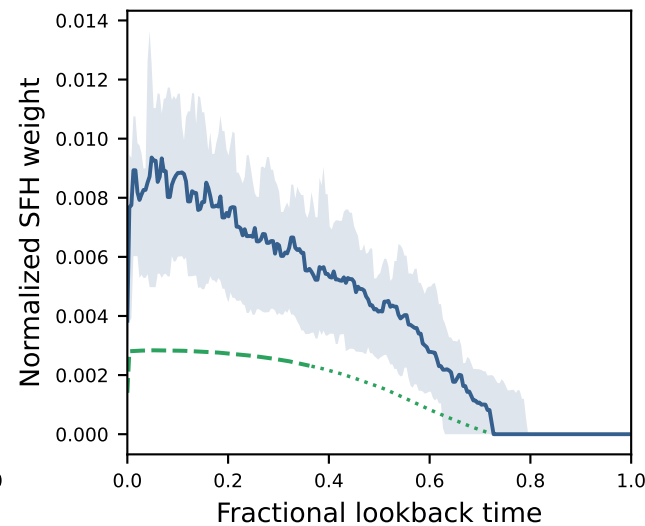
Complete candidate SFH



rank 5: 2737101536672032269
 $\log M^*=10.20$ (-0.10) $z=1.174$
 $\Delta MS=+1.67$ $D=0.448$

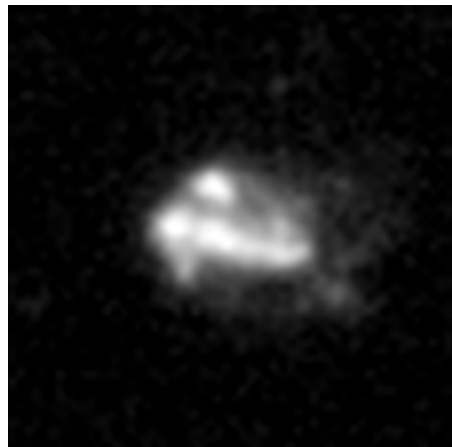


Complete candidate SFH

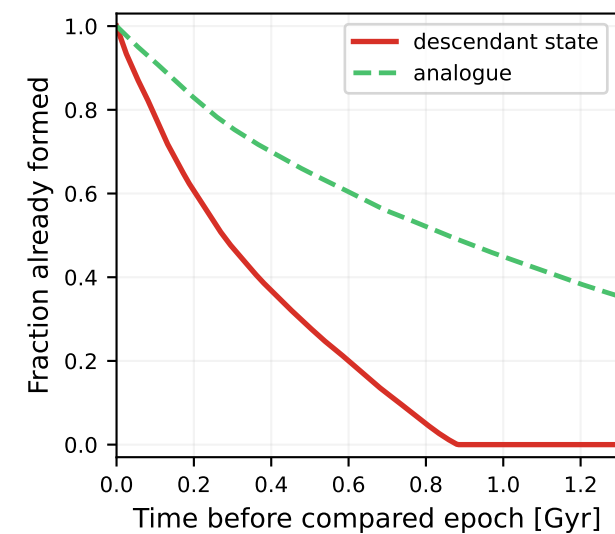
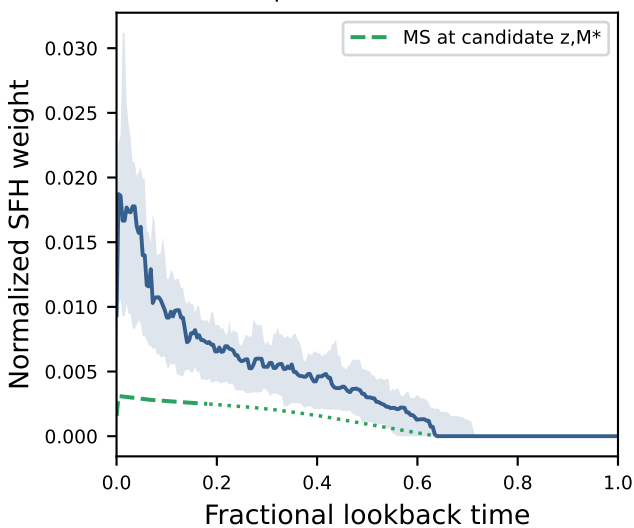


Earlier state $f=0.10$: lookback=5.37 Gyr, predicted $\log M^*=10.00$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

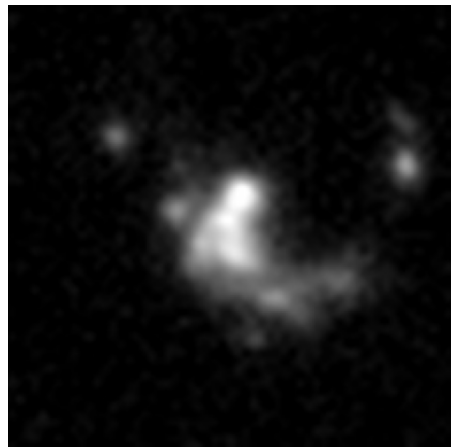
rank 1: 2647131718660160391
 $\log M^*=10.03 (+0.03)$ $z=1.170$
 $\Delta MS=+1.63$ $D=0.352$



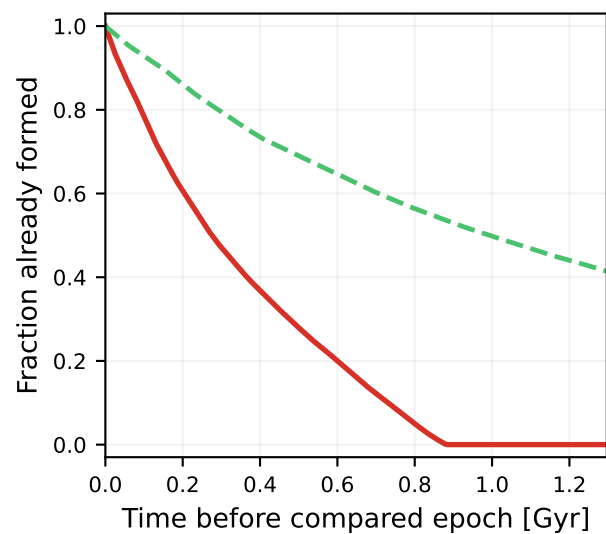
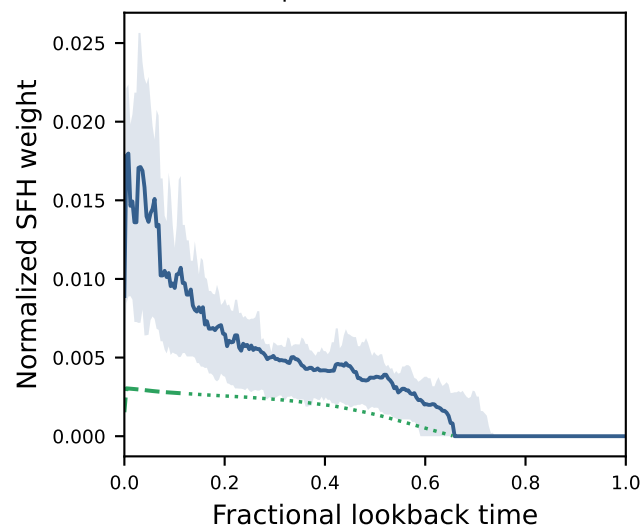
Complete candidate SFH



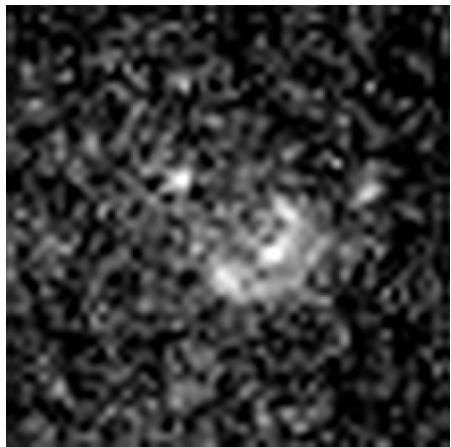
rank 2: 2650234495658951302
 $\log M^*=9.93 (-0.07)$ $z=1.010$
 $\Delta MS=+1.36$ $D=0.392$



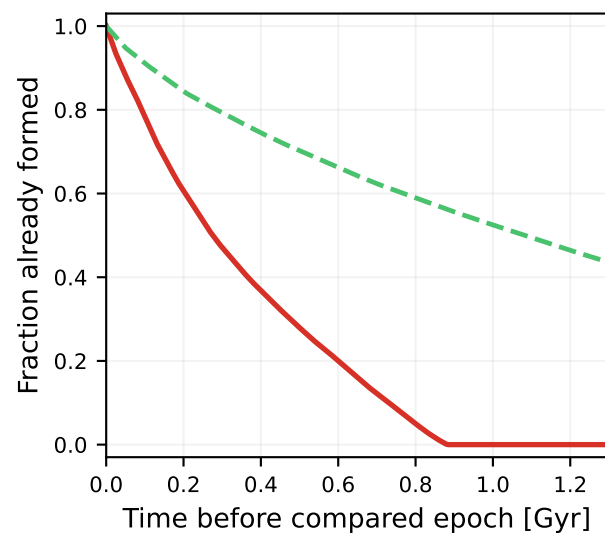
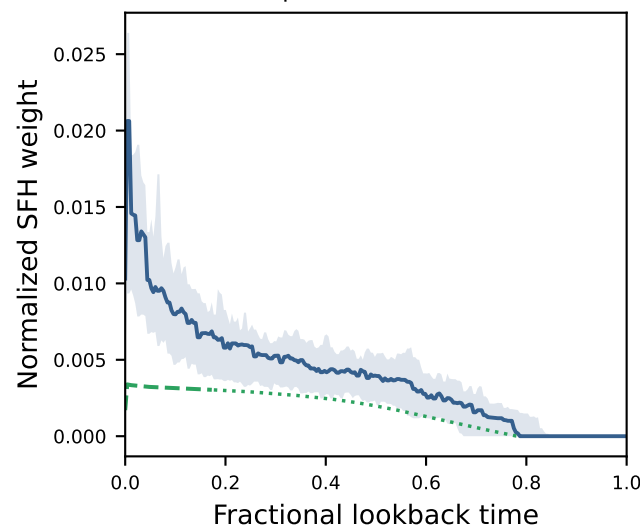
Complete candidate SFH



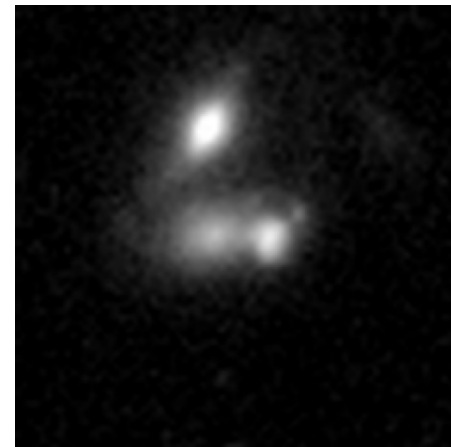
rank 3: 2647176063640000279
 $\log M^*=9.95 (-0.05)$ $z=1.240$
 $\Delta MS=+1.62$ $D=0.404$



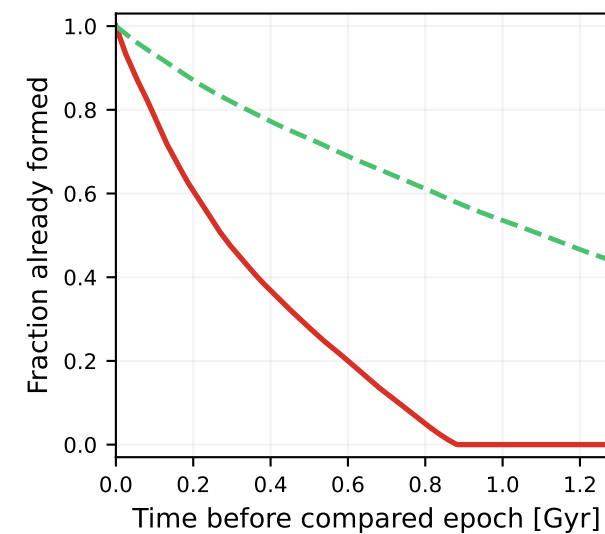
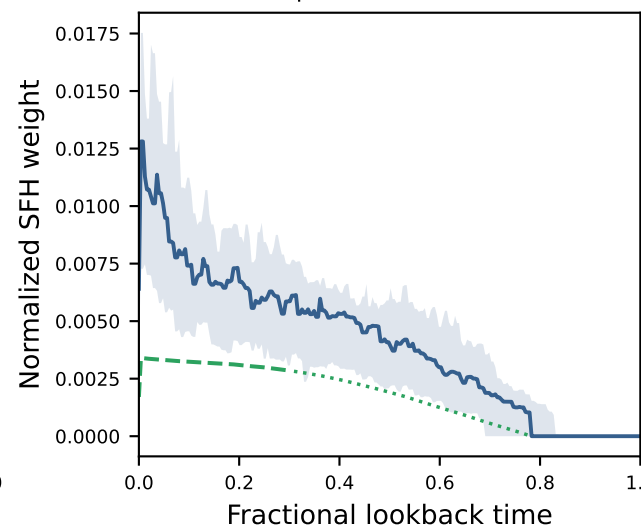
Complete candidate SFH



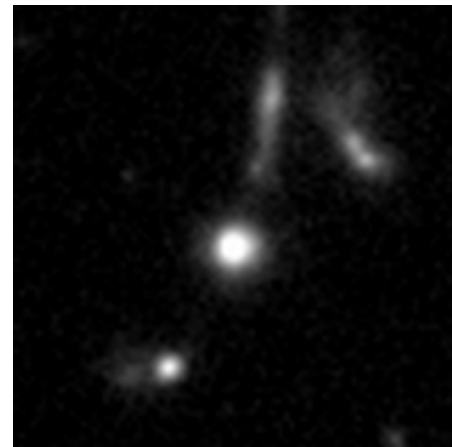
rank 4: 2658189408677769790
 $\log M^*=10.07 (+0.07)$ $z=1.464$
 $\Delta MS=+1.59$ $D=0.423$



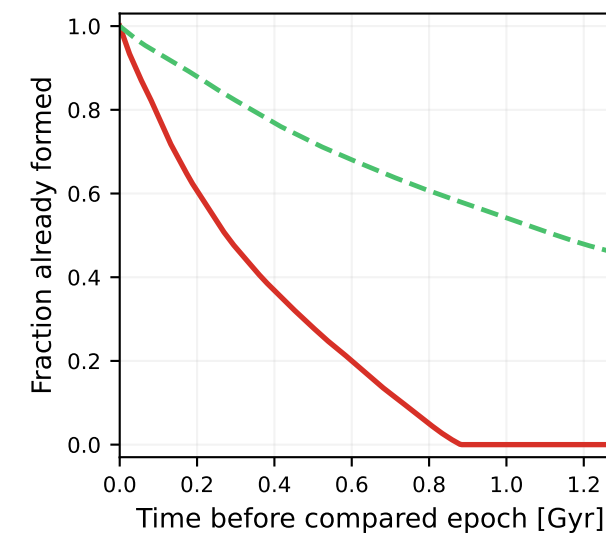
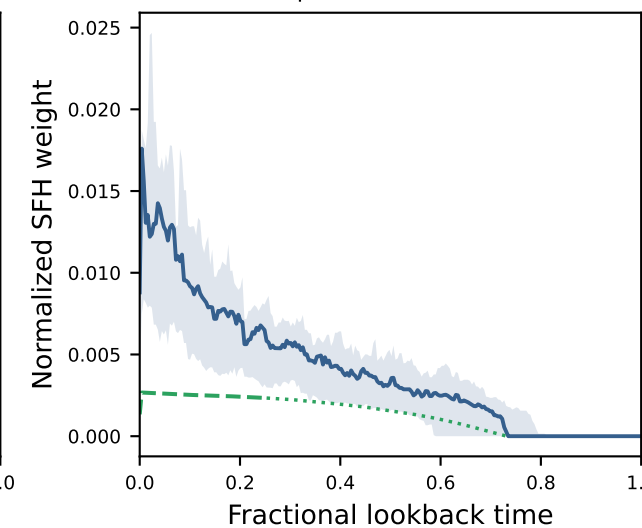
Complete candidate SFH



rank 5: 2684931694682266433
 $\log M^*=10.09 (+0.09)$ $z=0.984$
 $\Delta MS=+1.06$ $D=0.424$



Complete candidate SFH

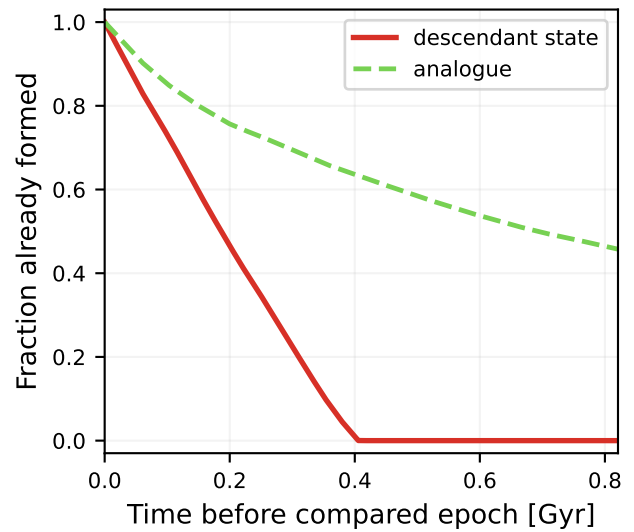
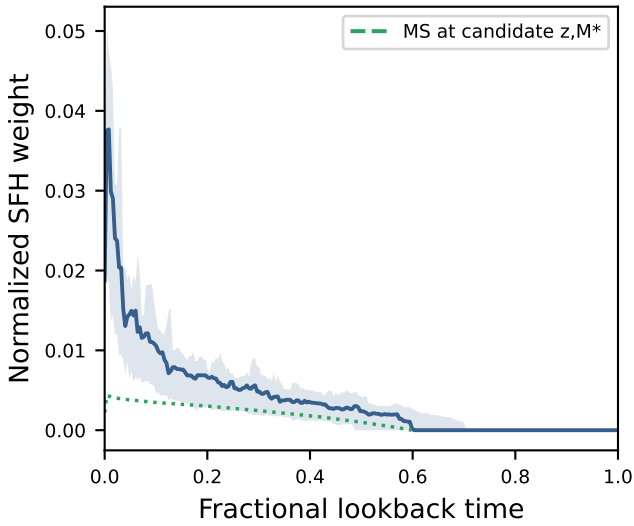


Earlier state $f=0.03$: lookback=5.85 Gyr, predicted $\log M^*=9.48$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

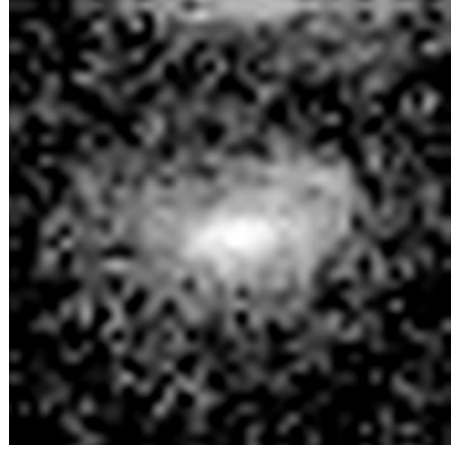
rank 1: 2648545772681764910
 $\log M^*=9.39$ (-0.08) $z=0.994$
 $\Delta MS=+1.79$ $D=0.421$



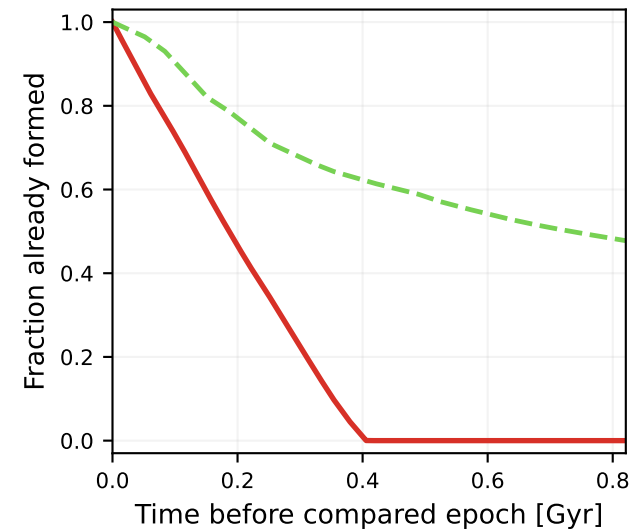
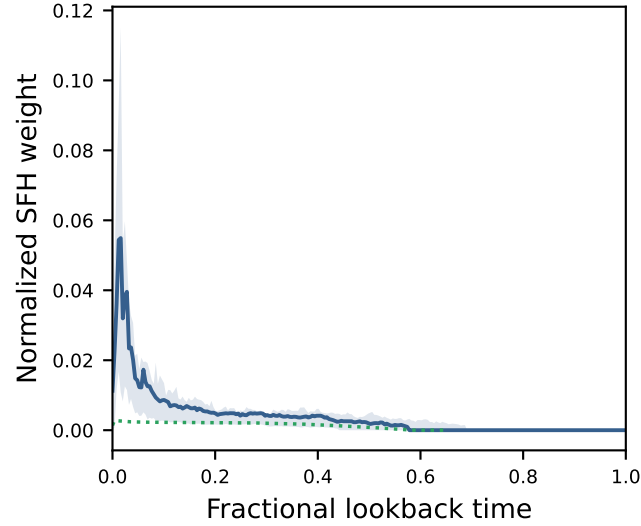
Complete candidate SFH



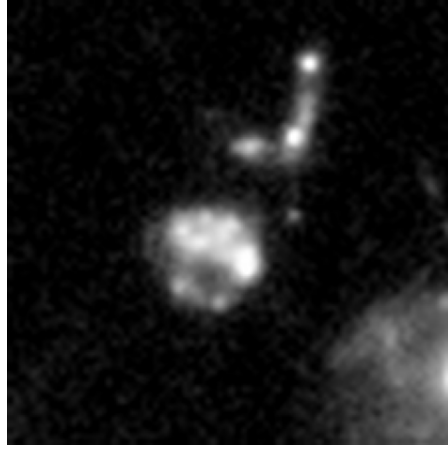
rank 2: 2647975483685285807
 $\log M^*=9.44$ (-0.04) $z=0.510$
 $\Delta MS=+1.34$ $D=0.429$



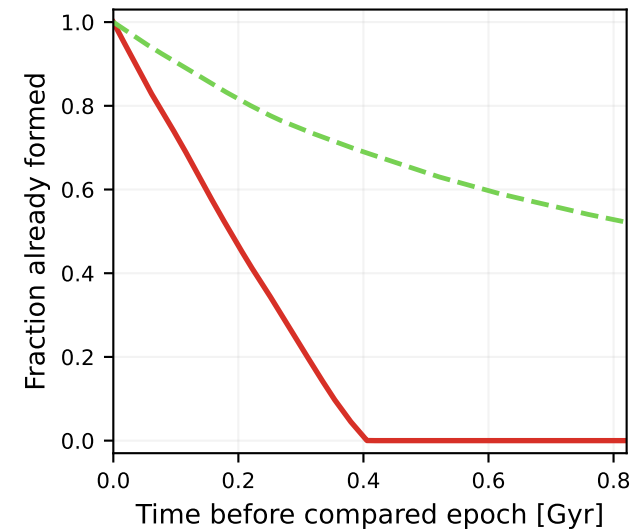
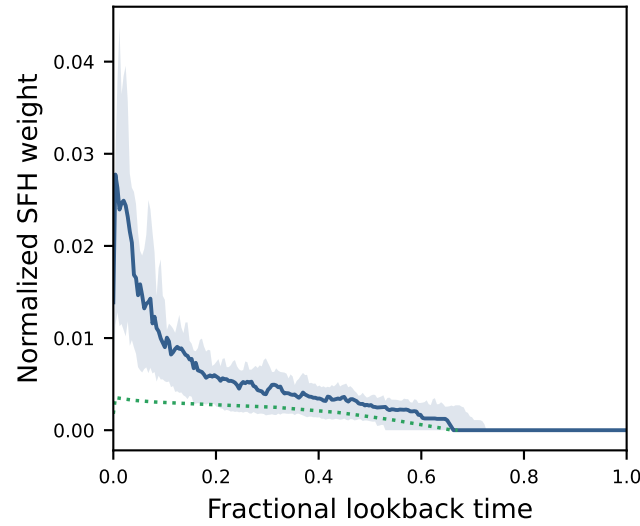
Complete candidate SFH



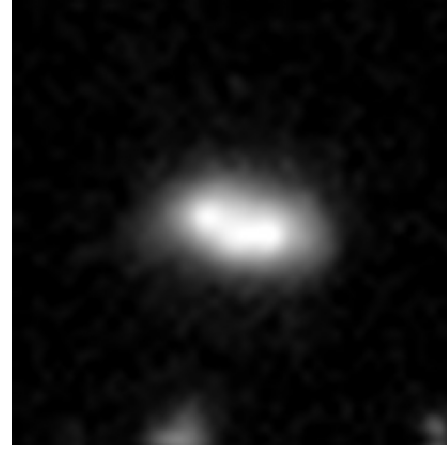
rank 3: 2740268308665100250
 $\log M^*=9.47$ (-0.01) $z=0.760$
 $\Delta MS=+1.77$ $D=0.475$



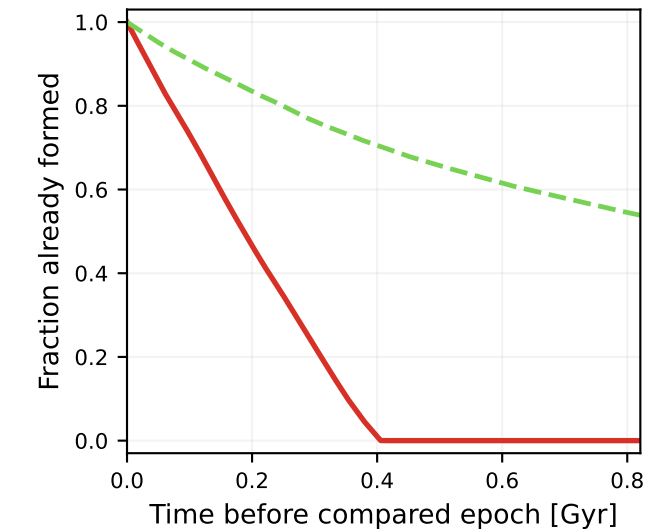
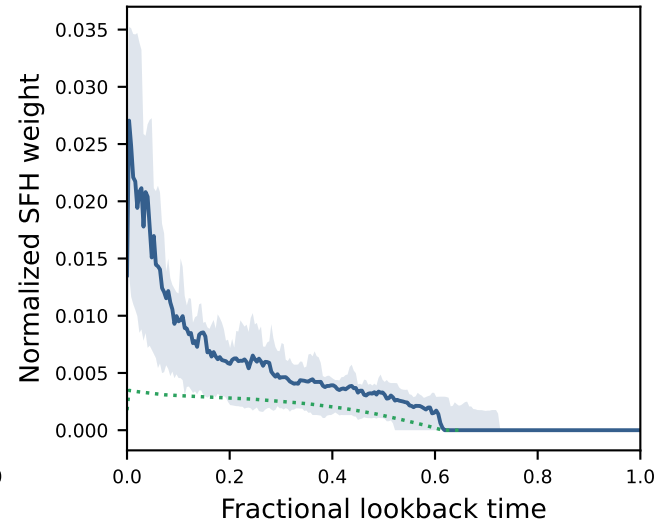
Complete candidate SFH



rank 4: 2729264667640595011
 $\log M^*=9.49$ (+0.01) $z=0.760$
 $\Delta MS=+1.78$ $D=0.490$



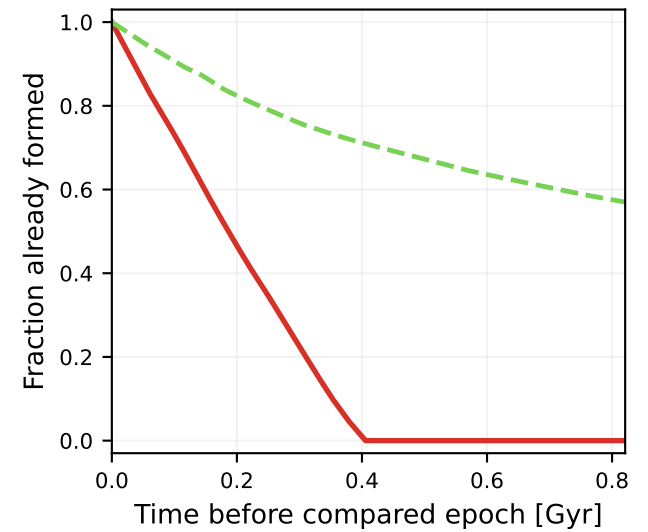
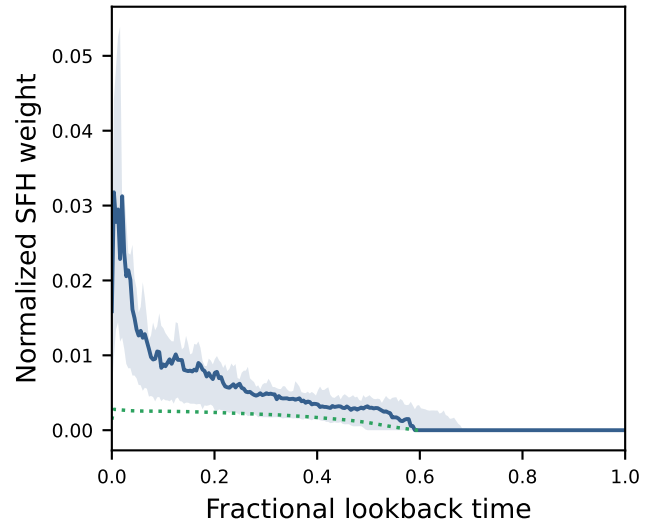
Complete candidate SFH



rank 5: 2753054265674679912
 $\log M^*=9.50$ (+0.02) $z=0.560$
 $\Delta MS=+1.19$ $D=0.499$

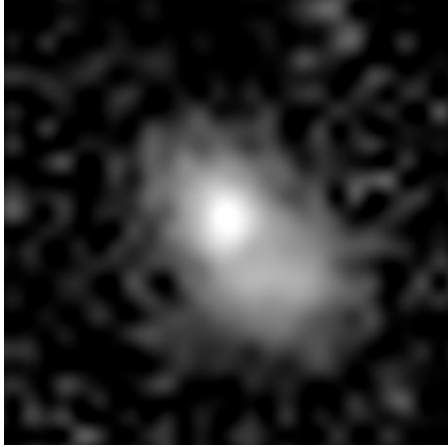


Complete candidate SFH

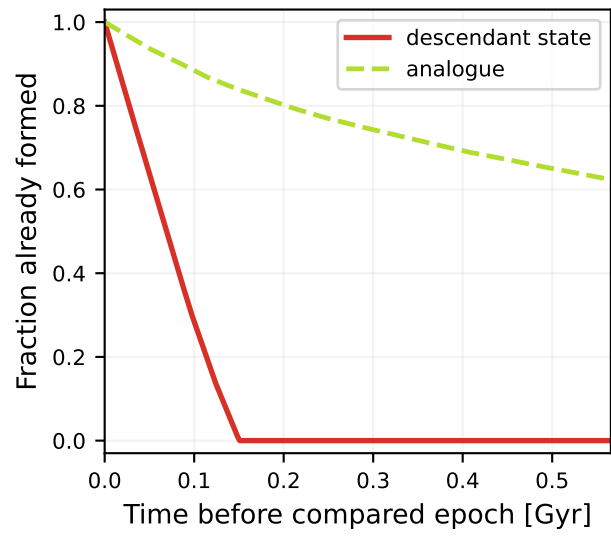
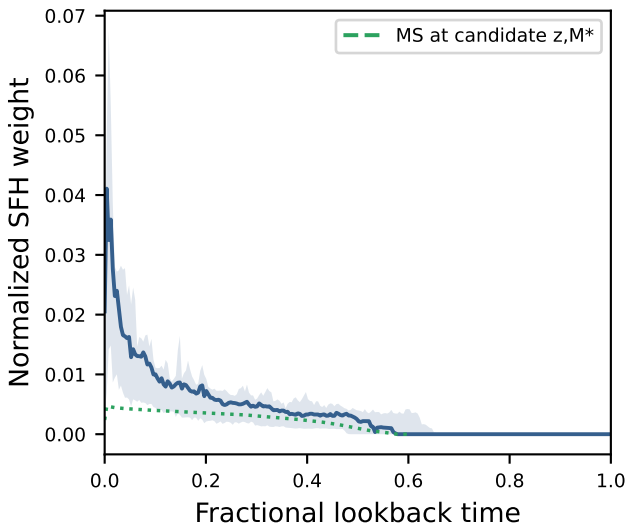


Earlier state $f=0.01$: lookback=6.10 Gyr, predicted $\log M^*=9.00$
 Candidates ranked by cumulative-SFH distance; redshift and morphology were not matched
 Green MS curves use each candidate's own observed z and mass ($R=0$; shifted MS)

rank 1: 2730218645688754837
 $\log M^*=8.88$ (-0.12) $z=0.560$
 $\Delta MS=+1.87$ $D=0.644$



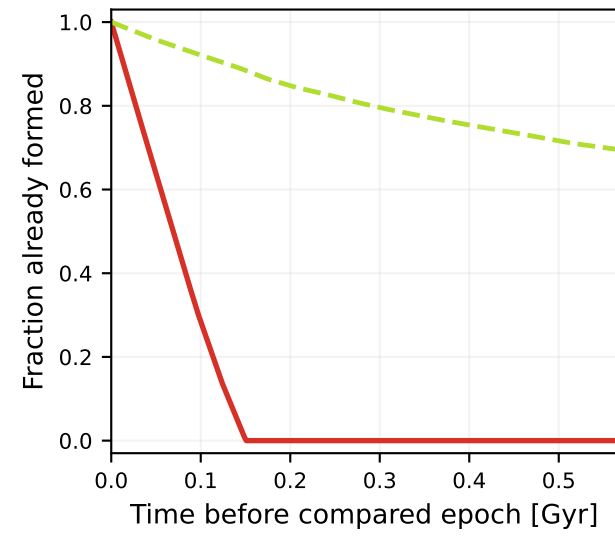
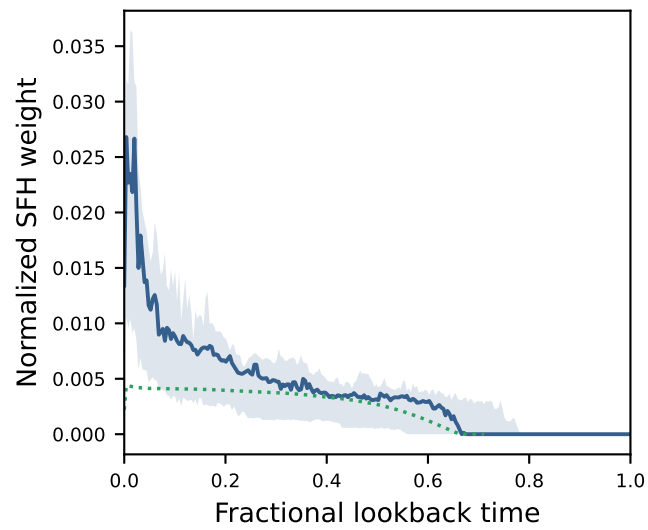
Complete candidate SFH



rank 2: 2634532333655237802
 $\log M^*=9.00$ (+0.00) $z=0.588$
 $\Delta MS=+1.86$ $D=0.694$



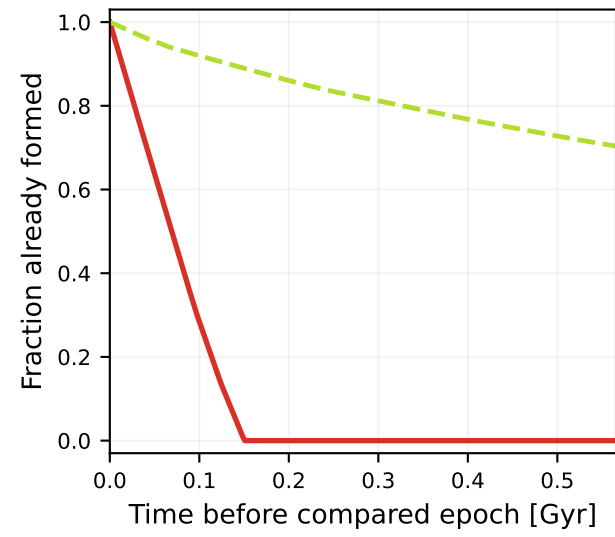
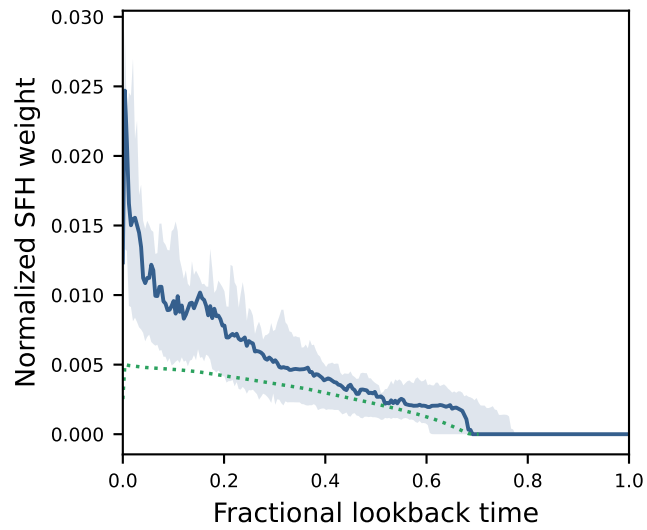
Complete candidate SFH



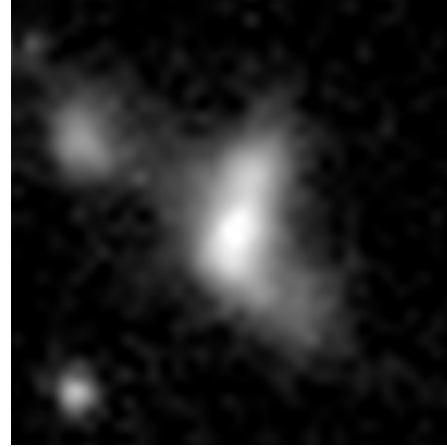
rank 3: 2674598393649348048
 $\log M^*=9.10$ (+0.10) $z=0.848$
 $\Delta MS=+1.80$ $D=0.703$



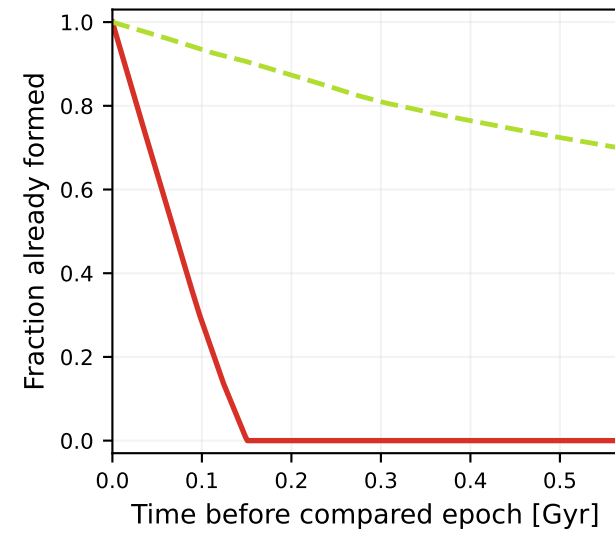
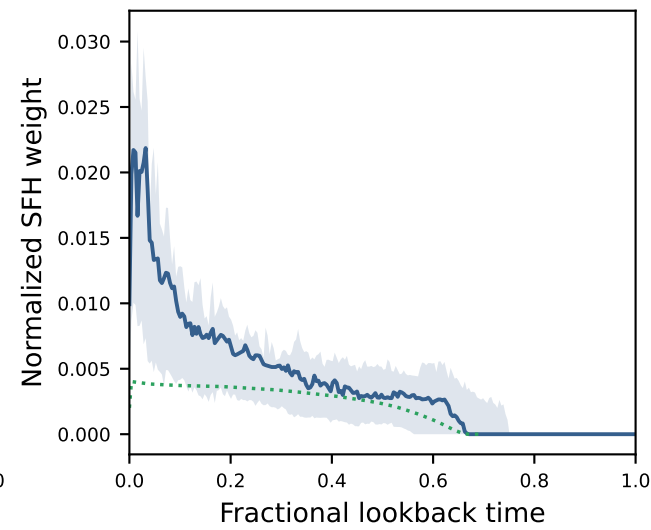
Complete candidate SFH



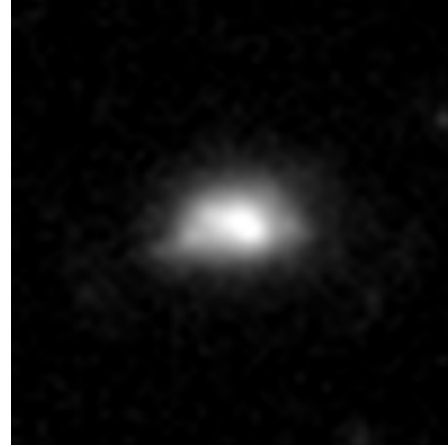
rank 4: 2765951524658897054
 $\log M^*=9.10$ (+0.10) $z=0.580$
 $\Delta MS=+1.72$ $D=0.707$



Complete candidate SFH



rank 5: 2662367379651485718
 $\log M^*=9.08$ (+0.08) $z=0.560$
 $\Delta MS=+1.53$ $D=0.725$



Complete candidate SFH

